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For correspondence:

johannes.schultz@ukbonn.de

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Contributions of insula and superior temporal sulcus to interpersonal guilt and responsibility in social decisions

Maria Gädeke¹, Tom Willems^{1,2}, Omar Salah Ahmed^{1,3,4}, Bernd Weber^{3,4}, René Hurlemann⁵,

Johannes Schultz^{3,4}✉

¹Masters in Neuroscience Program, University of Bonn, Bonn, Germany • ²Institute of Psychology, University of Bern, Bern, Switzerland • ³Center for Economics and Neuroscience, University of Bonn, Bonn, Germany • ⁴Institute of Experimental Epileptology and Cognition Research, Medical Faculty, University of Bonn, Bonn, Germany •

⁵Department of Psychiatry, University of Oldenburg, Oldenburg, Germany

eLife Assessment

This manuscript provides **valuable** novel insights into the role of interpersonal guilt in social decision-making by showing that responsibility for a partner's bad lottery outcomes influences happiness. Through the integration of neuroimaging and computational modelling methods, and by combining findings from two studies, the authors provide **solid** support for their claims. The findings will be of interest to researchers in the field of social neuroscience and decision making.

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Abstract

This study investigated the neural mechanisms involved in feelings of interpersonal guilt and responsibility evoked by social decisions. In two studies (one during fMRI), participants repeatedly chose between safe and risky monetary outcomes in social contexts. Across conditions, each participant chose for both themselves and a partner (*Social* condition), or the partner chose for both themselves and the participant (*Partner* condition), or the participant chose just for themselves (*Solo* condition, control). If the risky option was chosen in the *Social* or *Partner* condition, participant and partner could each receive either the high or the low outcome of a lottery with 50% probability, independently of each other. Participants were shown the outcomes for themselves and for their partner on each trial, and reported their momentary happiness every few trials. As expected, participant happiness decreased following both low lottery outcomes for themselves and for the partner. Crucially, happiness decreases following low outcomes for the partner were larger when the participant rather than their partner had made the choice, which fits an operational definition of guilt. This guilt effect was associated with BOLD signal increase in the left anterior insula. Connectivity between this region and the right inferior frontal gyrus varied depending on choice and experimental condition, suggesting that this part of prefrontal cortex is sensitive to guilt-related information during social choices. Variations in happiness were well explained by computational models based on participant's and partner's rewards and reward prediction errors. A model-based analysis revealed a left superior temporal sulcus cluster that tracked partner reward prediction errors that followed participant choices. Our findings identify neural mechanisms of guilt and social responsibility during social decisions under risk.

Introduction

Imagine that you go out to dinner with a friend and it is your turn to choose the restaurant. You can choose between two restaurants: one that you both know well, with very predictable food of good quality, and a new restaurant that neither you nor your friend has eaten in before. You

decide to try the new one. Unfortunately, while your dish is nice, your friend's turns out to be worse than the known restaurant's dishes. How would you feel? Would you feel differently if it had been your friend's turn to choose the restaurant?

Being responsible for such suboptimal outcomes for others can induce a feeling of interpersonal guilt, formally described as a negative emotional response to harming someone with whom one has a positive social bond (Baumeister, 1998 [Baumeister et al., 1994](#) [Berndsen et al., 2004](#) [Tangney et al., 2007](#) [Zeelenberg & Breugelmans, 2008](#)). Guilt influences decisions (e.g., Charness & Dufwenberg, 2006 [Tangney et al., 2007](#)), and abnormal sensitivity to guilt is associated with severe social dysfunctions ranging from psychopathy to depression and anxiety, depending on whether sensitivity to guilt is respectively reduced or increased ([Tangney et al., 2007](#)).

Several brain regions have been associated with the feeling of guilt: the anterior insula (aIns) (Bastin et al., 2016 [Lamm & Singer, 2010](#) [Piretti et al., 2023](#)), the dorsal cingulate cortex (Bastin et al., 2016 [Gifuni et al., 2017](#)) and the left temporo-parietal junction (Bastin et al., 2016 [Piretti et al., 2023](#)), the ventromedial prefrontal cortex (Krajbich et al., 2009 [Tangney et al., 2007](#)), and with various other regions and networks involved depending on the method used to induce guilt (Bastin et al., 2016 [Gifuni et al., 2017](#)). However, we are interested in the mechanisms involved in a specific, as yet understudied aspect of guilt: the kind that can result from everyday choices that expose others to an uncertain outcome. To study this, we built on findings from experimental social decision-making tasks.

Several studies have used games from behavioural economics or perceptual tasks linked to punishment of a partner. For example, participants behaving in accord with an economic definition of guilt aversion during a trust game showed activation in insula, supplementary motor area, dorsolateral prefrontal cortex (dlPFC), and temporal parietal junction (Chang et al., 2011 [Tangney et al., 2007](#)). In another study, partners received painful stimuli when participants made errors during a difficult perception task. These errors evoked activations in the left aIns and dlPFC in the participants (Koban et al., 2013 [Tangney et al., 2007](#)). In a variation of this task, participants could decide to bear a proportion of their partner's pain (Yu et al., 2014 [Tangney et al., 2007](#)). The level of pain taken, indicative of guilt, and activations in anterior middle cingulate cortex and aIns were higher when the pain followed errors made only by the participant rather than by both players. A multivariate reanalysis of these two datasets revealed a neural signature for guilt (Yu et al., 2020 [Tangney et al., 2007](#)), with key regions including the anterior medial cingulate cortex, insula, inferior frontal gyrus, inferior temporal cortex, thalamus and cerebellum.

One recent study paired the aforementioned perception-and-pain paradigm with dictator game decisions that allowed the participant to compensate for the partner's outcome (Gao et al., 2018 [Tangney et al., 2007](#)). The guilt context increased advantageous-inequity aversion and decreased disadvantageous-inequity aversion, and affected the neural correlates of these inequities (respectively: mentalizing-related, and emotion- and conflict-related regions). Finally, another study contrasted guilt and shame: confederates either experienced economic loss due to bad participant advice (participants experienced guilt), or experienced no loss when participants' bad advice was correctly refused by the confederate (participants experienced shame) (Zhu et al., 2019 [Tangney et al., 2007](#)). Guilt relative to shame activated supramarginal gyrus and temporo-parietal junction as well as orbitofrontal cortex, ventrolateral and dorsolateral prefrontal cortex. Multivariate analyses revealed that guilt could be distinguished from shame based on activation in ventral anterior cingulate cortex and dorsomedial prefrontal cortex.

One landmark study reported on the mechanisms involved in seeking or avoiding responsibility in decisions affecting a group of individuals, and reported involvement of medial prefrontal cortex, aIns and temporo-parietal junction (Edelson et al., 2018 [Tangney et al., 2007](#)). In this elegant study, participants could delegate their choice between a risky and a safe option to a group or decide themselves; in separate conditions, the choice affected payoff either only for the participant or for all group members. Interestingly, most participants displayed responsibility aversion, and this effect could not be explained by guilt, suggesting people associate a psychological cost with assuming responsibility for others' outcomes. However, in many situations one does not have the possibility to delegate a decision to others, and choosing a risky option in such a case may evoke guilt when

the outcome for others is negative. The neural mechanisms underlying guilt evoked during such situations of social responsibility are still unknown. These questions are subject of the present study. Similarly to Edelson et al.'s and more recent studies (Arioli et al., 2023 [↗](#); Fareri et al., 2022 [↗](#)), our paradigm leveraged a risky choice task with social conditions.

In our experiment, participants were paired with a partner and played an “ice-breaker” game that created a positive social bond between them, increasing the likelihood of feeling empathy and guilt for each other (Baumeister, 1998 [↗](#); Julle-Danière et al., 2020 [↗](#); Loewenstein et al., 1989 [↗](#)). Participants or their partner then chose between risky and safe monetary options in three conditions (Figure 1 [↗](#)). In the *Social* condition, the outcome of choices affected both participant and partner; participants were and felt responsible for these outcomes because they had agency over their decisions and knew that they could have chosen otherwise (Frith, 2014 [↗](#)). We contrasted this condition with similar choices made by a simple expected-value-maximizing algorithm posing for the partner (*Partner* condition), and participant choices only affecting the participant themselves (*Solo* condition, acting as control). Importantly, we assessed the emotional impact of the outcomes of these choices by monitoring participants' happiness every two trials (Rutledge et al., 2014 [↗](#), 2016 [↗](#)). This allowed to fit computational models to happiness data and search for networks sensitive to reward prediction errors resulting from participant or partner choices. We ran two experiments, Study 1 outside the MRI scanner and Study 2 during fMRI, with separate groups of participants.

We analysed our behavioural data using several complementary methods: choices were modelled with mixed-effects regressions serving as manipulation checks; risk preferences expressed in choices were assessed using a comprehensive expected utility model as well as with a simpler, more robust “risk premium” approach; and happiness data were fitted, in addition to the computational models, with several linear mixed models to assess the impact of both the participant's and their partner's rewards, the impact of agency and their interactions. Inspired by findings reported in previous neuroimaging of social emotions, we also used several methods to analyse our fMRI data, including conventional methods (both region-of-interest and mass univariate); mixed-effects regression models; computational model-based analyses (inspired by e.g. Konovalov et al., 2021 [↗](#); Rutledge et al., 2014 [↗](#)); and functional connectivity (e.g. Edelson et al., 2018 [↗](#); Konovalov et al., 2021 [↗](#)). The behavioural modelling is thus complemented by neuroimaging analyses that offer insight about both the activity in regions associated with guilt as well as their place in a wider network, providing an in-depth comprehensive analysis of the mechanisms behind guilt evoked by social responsibility.

Results

Behaviour

Overview

We first verified that participants' choices were reasonable in the *Solo* and *Social* conditions, then assessed whether these choices and the risk preferences they revealed changed depending on whether participants chose just for themselves (*solo*) or for themselves and the interaction partner (*social*). Next, we assessed whether momentary happiness varied with decision outcomes and whether responsibility influenced this relationship. Lastly, we fitted computational models to the happiness data. While Study 1 (behaviour only) was run before Study 2 (fMRI), we will report the results of both studies together as their results were highly consistent.

Choices: manipulation check

As expected, participants' probability of choosing the risky option (lottery) increased with the difference between the expected value of the lottery and the value of the safe option (Study 1: Figure 2A [↗](#), $t(4796) = 9.26$, $p < 3.1e^{-20}$, $\beta = 0.074$, 95% $CI = [0.059\ 0.090]$; Study 2: Figure 2D [↗](#), $t(3829) = 10.62$, $p < 5.3e^{-26}$, $\beta = 0.093$, 95% $CI = [0.075\ 0.110]$; mixed-effects regressions, see Equation 1 [↗](#), detailed results are reported in Supplementary Table 1 [↗](#)). Participants chose the lottery more often in the *Solo* condition than in the *Social* condition in Study 1 ($t(4796) = 2.54$, $p = 0.011$, $\beta =$

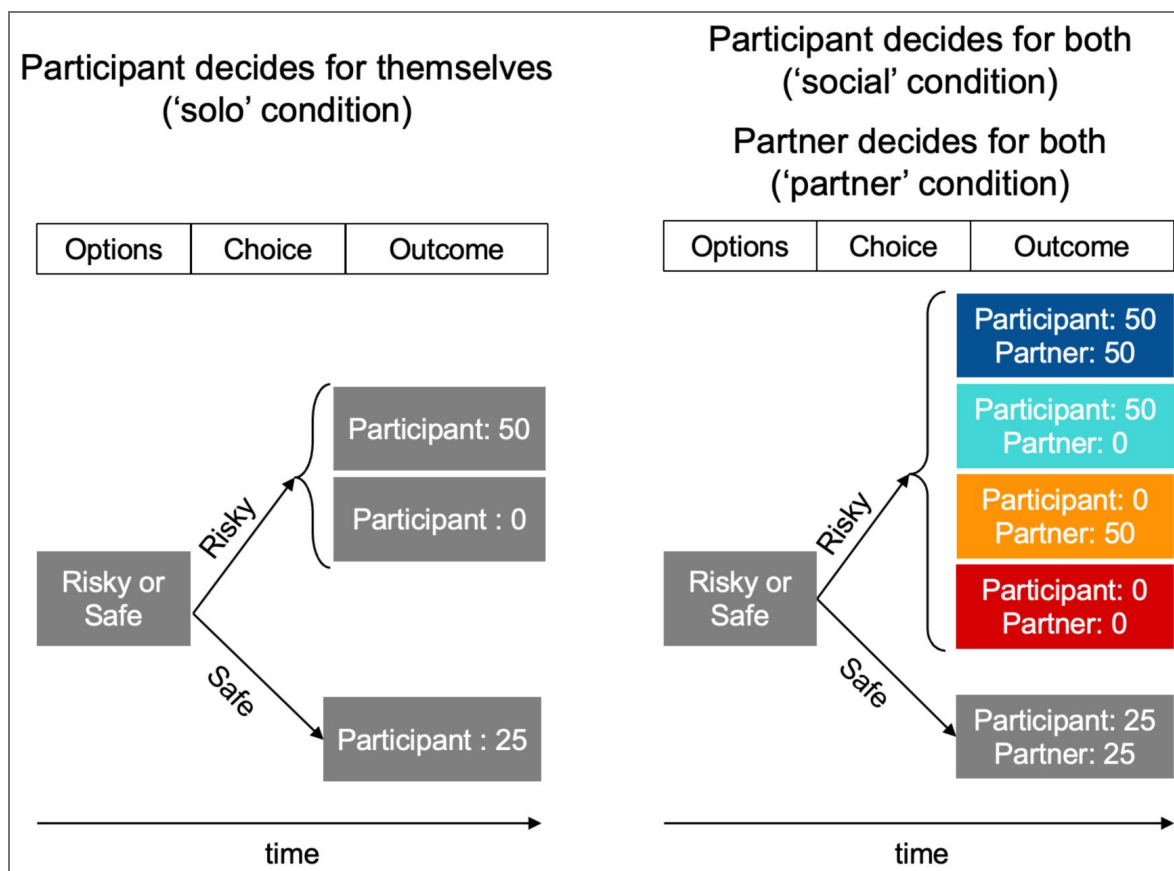


Figure 1. Experimental design.

In every trial, participants were presented with pairs of monetary options (a safe and a risky option; the risky option was a lottery with equally probable high and low outcomes). There were three conditions: a non-social 'Solo' condition, in which the participant's choice led to an outcome just for themselves (left panel); and two conditions in which choices were made by the participant ('Social' condition) or by their partner ('Partner' condition) and lead to outcomes affecting both players (right panel). Importantly, selecting the risky option in the social or partner conditions lead to the lottery being played out independently for both players; i.e., participant and partner could receive the high or low outcome independently from each other (coloured boxes). Selecting the safe option led to both players receiving an equal outcome.

0.164, 95% $CI = [0.038\ 0.291]$), but this difference was not found in Study 2 ($t(3829) = 0.23$, $p = 0.82$, $\beta = 0.015$, 95% $CI = [-0.109\ 0.138]$). There was no significant interaction between the difference in expected values and experimental conditions in either study ($p > 0.52$).

Choices: risk preferences

To better assess whether people's risk preferences varied between the *Solo* and the *Social* condition, we evaluated two additional measures. First, we calculated for each participant a "risk premium", defined as the difference between the expected value of a lottery and its certainty equivalent (See Equation 2 [in Methods](#); positive risk premiums indicate risk aversion).

Second, we used an expected utility theory (EUT) approach to calculate a parameter ρ that describes a decision-maker's risk attitude under the assumption of constant absolute risk aversion (see Equation 3 [in Methods](#)). We used two measures because fitting the EUT model, the more comprehensive measure that takes into account all choices of a participant, requires many trials to be fitted reliably, a condition that was not satisfied in many participants of both studies; in contrast, the risk premium's single-point measure of risk aversion could be estimated in all participants. Risk premiums did not differ between *Social* and *Solo* conditions (Study 1: Figure 2B [in Results](#), $t(39) = 1.53$, $p = 0.134$, $Cohen's\ d = 0.24$, $BF_{10} = 0.49$; Study 2: Figure 2E [in Results](#), $t(43) = -0.21$, $p = 0.84$, $d = -0.03$, $BF_{10} = 0.17$). Turning to the EUT approach, as differences in risk attitudes for gains and losses are well known (Kahneman & Tversky, 1979 [in Methods](#)), we first estimated ρ separately for gain and loss trials. As the number of these types of trials varied across participants, we only obtained reliable estimates for both types of trials in some participants (Study 1: 18 of 40; Study 2: 16 of 44). As ρ did not vary between gain and loss trials (Study 1: $t(17) = 0.21$, $p = 0.84$, $d = 0.05$; Study 2: $t(15) = -0.61$, $p = 0.55$, $d = 0.15$; paired t-test), we then pooled across gain and loss trials, then estimated and compared ρ for the *Solo* and *Social* conditions using paired t-tests. We found that participants were slightly more risk averse in the *Social* than in the *Solo* condition in Study 1 (Figure 2C [in Results](#), $t(39) = 2.27$, $p = 0.03$, $d = 0.36$, $BF_{10} = 1.69$) but not in Study 2 (Figure 2F [in Results](#), $t(43) = 1.40$, $p = 0.17$, $d = 0.21$, $BF_{10} = 0.41$). Risk premium and ρ were highly consistent with each other across participants in both conditions for both studies (Study 1: *Solo* condition: $F(1,38) = 69.9$, $p < 0.001$, $R^2 = 0.65$; *Social* condition: $F(1,38) = 42.0$, $p < 0.001$, $R^2 = 0.55$; Study 2: *Solo* condition: $F(1,41) = 57.9$, $p < 0.001$, $R^2 = 0.59$; *Social* condition: $F(1,41) = 74.2$, $p < 0.001$, $R^2 = 0.64$; these and all subsequent regressions are robust).

In sum, participants showed very similar risk preferences when making decisions affecting only themselves (*Solo* condition) or themselves and their partner (*Social* condition), with a tendency towards higher risk aversion in the *Social* condition in Study 1.

Momentary happiness: links to reward

Momentary happiness was assessed every two trials in a similar manner to previous studies by Rutledge and colleagues (Rutledge et al., 2014 [in Methods](#), 2016 [in Methods](#)). Across all trials, in both studies, participant momentary happiness correlated with rewards obtained in the current trial by the participant and by the partner (Correlations with participant reward, Study 1: $F(1,3598) = 691.5$, $p < 0.001$, $R^2 = 0.16$; partner reward, Study 1: $F(1,2426) = 128.6$, $p < 0.001$, $R^2 = 0.05$; participant reward, Study 2: $F(1,2630) = 650.8$, $p < 0.001$, $R^2 = 0.20$; partner reward, Study 2: $F(1,1735) = 111.8$, $p < 0.001$, $R^2 = 0.06$; Figure 3A [in Results](#), 3B [in Results](#), 3E [in Results](#) and 3F [in Results](#)). Based on previous findings (Rutledge et al., 2016 [in Methods](#)), we reasoned that participants' momentary happiness would be influenced not only by rewards obtained in the current trial, but also by expected rewards, reward prediction errors, rewards received in previous trials, and differences between rewards obtained by the participant and the partner. Crucial for our research question, we aimed to assess whether responsibility for these rewards, i.e., taking into account whether the rewards occurred following choices made by the participant or the partner, would also influence variations in happiness.

Following Rutledge and colleagues' methodology, which considers that changes in momentary happiness in response to outcomes of a probabilistic reward task are explained by the combined influence of recent reward expectations and prediction errors arising from those expectations, we fitted computational models to each participant's happiness data. In these models, certain rewards, the expected value of chosen lotteries, reward prediction errors and additional outcome

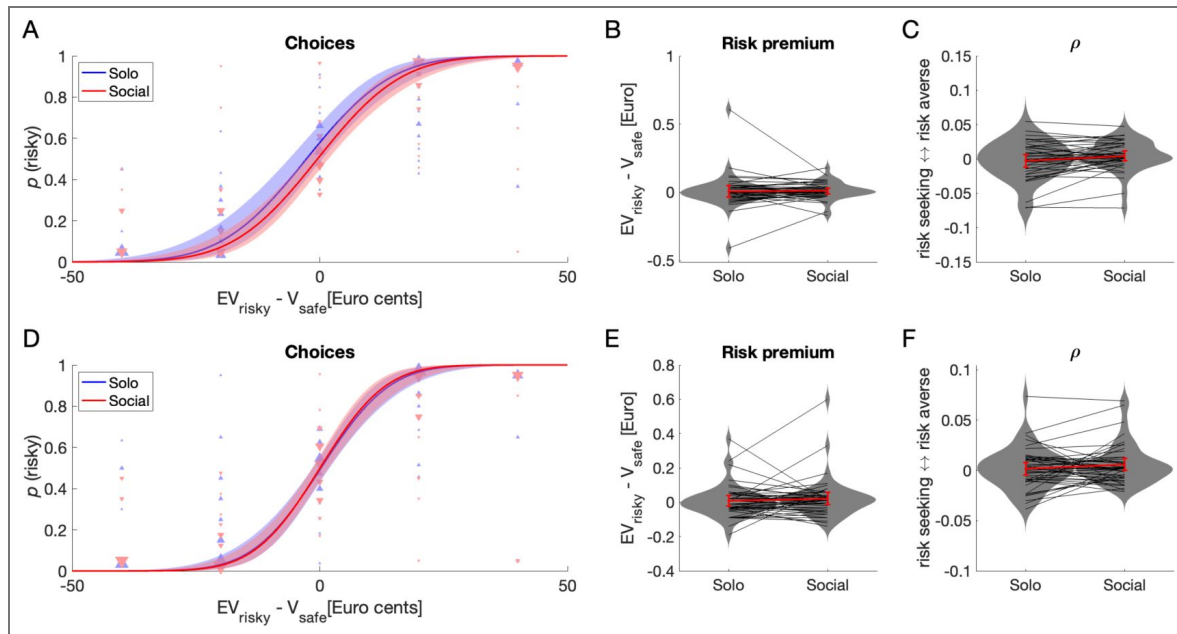


Figure 2. Participant choices in Studies 1 (outside fMRI, N = 40) and 2 (inside fMRI, N = 44).

A and **D**. The probability of choosing the risky option (lottery) in both Solo and Social conditions is well explained by the difference in expected value of the risky and safe choice options ($EV_{\text{risky}} - V_{\text{safe}}$). Participants chose the risky option slightly more often in the Solo condition than in the Social condition in Study 1 (**A**) but not in Study 2 (**D**). Lines are predicted values of a logit linear mixed regression model fitted to the choice data (see Results). Error areas indicate 95% pointwise confidence intervals for the predicted values. Triangles indicate individual average choice proportions binned by $EV_{\text{risky}} - V_{\text{safe}}$ value; size of triangle reflects the number of participants contributing to a datapoint. Blue upwards-pointing and red downwards-pointing triangles are data from the Solo and Social conditions, respectively. **B** and **E**. Risk premiums did not differ between Solo and Social conditions. **C** and **F**. Values of the risk aversion parameter ρ in the Solo and Social conditions were broadly consistent with Risk premium values, but showed that participants were slightly more risk averse in the Social than in the Solo condition in Study 1 only (see Results). In panels B, C, E and F, grey lines and markers show individual data, red lines show means with 95% confidence intervals about the means, and grey areas are kernel density plots representing the distribution of the data.

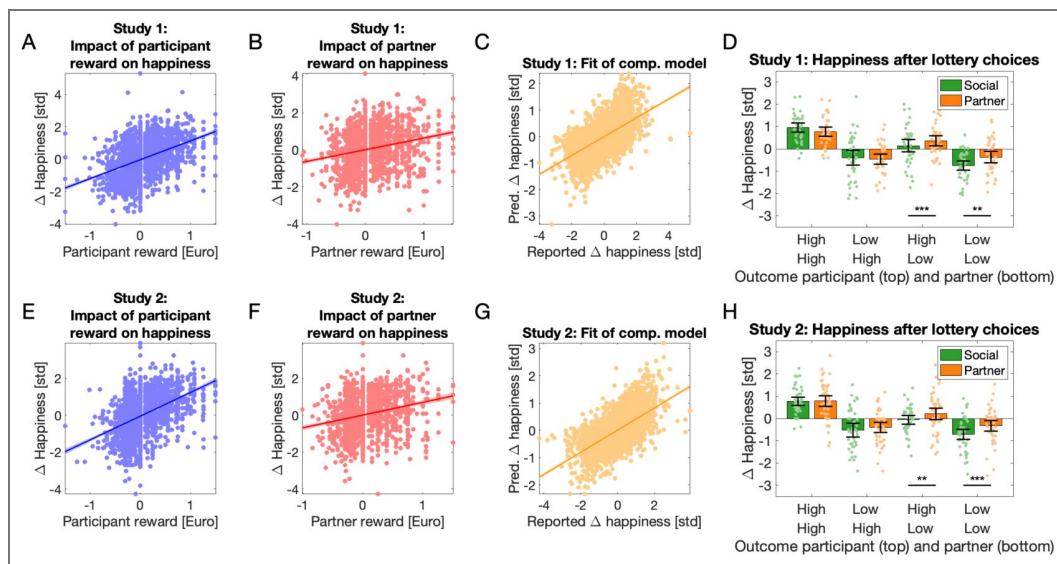


Figure 3. Participant momentary happiness in Studies 1(A-D) and 2 (E-H).

Happiness varied with rewards received by the participant (**A** and **E**) and by the partner (**B** and **F**). Each dot is one trial; data are pooled across participants. Lines are fitted regression lines. A computational model taking into account expected, previous and current rewards, reward prediction errors for both participant and partner, and decision-maker (Responsibility Redux model, see Results) predicted the variations in participants' momentary happiness well (**C** and **G**, and Table 1 [link](#)). Changes in momentary happiness after lottery choices in Social and Partner conditions varied with lottery outcome and decision-maker (**D** and **H**). Data were binned according to outcome for each participant and decision-maker (Social = participant chose the lottery, Partner = partner chose the lottery). Crucially, responsibility for low lottery outcomes for the partner decreased participant happiness more than the same outcomes following partner choices (see Results), which fits the definition of interpersonal guilt. In **D** and **H**, dots are individual datapoints, the bar indicates the mean, the error bars are 95% confidence intervals about the mean, and the stars indicate the significance of the "guilt effect" (see text): *** $p < 0.001$; ** $p < 0.01$.

parameters are modelled separately with influences that decay exponentially with time (see Methods for equations and details). We first fitted a ‘Basic’ model (Equation 4) containing regressors coding for certain reward, expected value of chosen lotteries and participant reward prediction errors to evaluate whether these fundamental reward variables explained variations in momentary happiness. This model explained the data reasonably well (see Table 1). Next, based on the notion that inequality between rewards received by the participant and the partner would influence happiness (e.g., Loewenstein et al., 1989), we assessed the fit of an ‘Inequality’ model that included one term modelling the difference between the participant and partner outcomes (Equation 5). We also assessed the fit of Rutledge and colleagues’ (2016) ‘Guilt-envy’ model in which advantageous and disadvantageous inequality were modelled separately (Equation 6). Next, we tested a ‘Responsibility’ model in which separate terms represented the partner’s reward prediction errors resulting from choices made by the participant and by the partner (Equation 7). Finally, we optimised the Responsibility model by removing the regressor modelling partner reward prediction errors resulting from choices made by the partner, yielding a ‘Responsibility Redux’ model (Equation 8).

All models explained variations in happiness reasonably well (Table 1 and Supplementary Materials). Overall, the *Responsibility* and *Responsibility Redux* (Figure 3C and 3G) models explained the data best (highest R^2 / adjusted R^2 or lowest AIC / BIC); a likelihood ratio test (Equation 9) revealed that the *Responsibility* model fitted better than all the other models, including the *Responsibility Redux* model (Study 1: all $LR \geq 47.36$, $p < 0.0001$; Study 2: all $LR \geq 77.83$, $p < 0.0001$). Weights for certain rewards, expected value of the lotteries, participant reward prediction errors and the forgetting factor γ were overall positive across participants, models and studies (Study 1: all $Z > 5.4$, $p < 0.0001$; Study 2: all $Z > 4.9$, $p < 0.0001$, Wilcoxon sign-rank tests were used because data were not normally distributed). Medians of the forgetting factor γ varied little across models, ranging from 0.39 to 0.44 in Study 1 and from 0.42 to 0.46 in Study 2, indicating stable influences of previous trials on happiness across models. Participants’ reward prediction errors (sRPE) influenced happiness more than partner’s reward prediction errors, whether the latter resulted from the participant or the partner’s choices (respectively *social_pRPE* and *partner_pRPE*): weights for sRPE were higher than for *social_pRPE* or *partner_pRPE* (Study 1: all $Z > 6.0$, $p < 0.001$; Study 2: all $Z > 3.7$, $p < 0.003$). The stability of these estimated parameters was verified using a parameter recovery procedure (see Methods and Supplementary Materials). These results thus replicate the finding that outcomes of risky social decisions influence momentary happiness (Rutledge et al., 2016). Crucially, we find here that the partner’s reward prediction errors (*social_pRPE* and *partner_pRPE*) contributed to explaining changes in participants’ momentary happiness: the *Responsibility* and *Responsibility Redux* models explained the data better than the models without these parameters (see Table 1). In particular, the partner’s reward prediction errors resulting from the participants’ decisions (*social_pRPE*), i.e. those pRPE for which participants were responsible, contributed to explaining our data (weights for *social_pRPE* were greater than 0: *Responsibility model*: Study 1: $Z = 2.85$, $p = 0.004$, Study 2: $Z = 3.26$, $p = 0.001$; *Responsibility Redux model*: Study 1: $Z = 2.93$, $p = 0.003$, Study 2: $Z = 3.30$, $p = 0.001$; weights for *social_pRPE* tended to be higher than weights for *partner_pRPE*: *Responsibility model*: Study 1: $Z = 2.14$, $p = 0.033$; Study 2: $Z = 1.41$, $p = 0.16$).

Momentary happiness: effects of agency, responsibility and guilt

Next, we assessed whether happiness varied depending on the participant’s agency (*Social* + *Solo* vs. *Partner*), and found happiness to be lower when the participant chose, independent of the outcome (Study 1: $t(3600) = -3.92$, $p < 0.0001$, $\beta = -0.14$, 95% CI = [-0.20 -0.07]; Study 2: $t(2870) = -6.07$, $p < 0.0001$, $\beta = -0.24$, 95% CI = [-0.31 -0.16]). This is interesting in itself and may reflect the drive behind responsibility aversion reported by Edelson et al.’s 2018 study: being assigned the role of the decider in a social setting may make people slightly unhappy, perhaps due to “weight of the responsibility”. To specifically search for a sign of interpersonal guilt, we analysed happiness values reported after outcomes of lottery choices in the *Social* and *Partner* conditions using conventional linear mixed models (Equation 10). Fitting several models revealed that lottery outcomes influenced happiness, and that the identity of the decision-maker played a role

<i>Model</i>	<i>N param</i>	<i>Mean R^2</i>	<i>Mean R^2_{adj}</i>	<i>BIC</i>	<i>AIC</i>
<u>Study 1</u>					
Basic	3	0.328	0.305	-1000	-1399
Inequality	4	0.346	0.316	-916	-1416
Guilt-envy	5	0.354	0.316	-785	-1385
Responsibility	5	0.370	0.333	-866	-1466
Responsibility Redux	4	0.361	0.331	-999	-1499
<u>Study 2</u>					
Basic	3	0.374	0.340	-693	-1062
Inequality	4	0.394	0.350	-620	-1080
Guilt-envy	5	0.405	0.350	-491	-1043
Responsibility	5	0.433	0.380	-616	-1168
Responsibility Redux	4	0.422	0.379	-735	-1195

BIC, Bayesian Information Criterion. AIC, Akaike's Information Criterion. BIC and AIC values are summed across participants. As in previous work by Rutledge and colleagues (Rutledge et al., 2014, 2016), model fits were performed with individually z-scored happiness ratings. Best values of each variable for each study are highlighted in bold font. For model details, see Results and Methods (Equations 4 to 8).

Table 1. Fits of computational models to momentary happiness data.

(Supplementary Table 2 [↗](#)). Crucially, the interaction between partner outcome and decision-maker was significant (Study 1: $t(1180) = 3.52$, $p = 0.0004$, $\beta = 0.37$, 95% CI = [0.16 0.58]; Study 2: $t(937) = 2.85$, $p = 0.0045$, $\beta = 0.33$, 95% CI = [0.10 0.56]).

When the partner received the low lottery outcome, participant happiness was lower when they rather than the partner had chosen the lottery (Figure 3D [↗](#) and 3H [↗](#)). When the partner received the low lottery outcome of a participant-chosen lottery, they would presumably feel let down because the participant could have chosen the better safe option. As participants knew this, they were likely to feel “simple guilt” (Battigalli & Dufwenberg, 2007 [↗](#)).

This behavioural effect (difference in happiness obtained when the partner received low lottery outcomes after participant rather than partner choices) is thus compatible with “simple guilt”, and we will thus refer to it as “guilt effect”. The “guilt effect” occurred whether the participant received the high lottery outcome (Study 1: $t(39) = -3.58$, $p < 0.001$, $d = 0.56$, $BF_{10} = 32$; Study 2: $t(43) = -2.68$, $p = 0.01$, $d = 0.4$, $BF_{10} = 3.8$) or the low lottery outcome (Study 1: $t(39) = -3.39$, $p = 0.002$, $d = 0.54$, $BF_{10} = 19$; Study 2: $t(43) = -3.58$, $p < 0.001$, $d = 0.54$, $BF_{10} = 33.5$). In Study 2, we will compare individual “guilt effect” values to individual guilt-related brain activation patterns (see end of the section on BOLD signal results). Responsibility for choices did not influence happiness following positive lottery outcomes for the partner (both studies, all $|t| < 1.3$, $p > 0.2$, $BF_{10} < 0.2$).

In sum, in both studies, we found evidence that participants’ happiness was influenced by the outcomes of choices for their partner, especially by outcomes resulting from participant decisions, i.e., partner outcomes for which participants were responsible. Within these outcomes, participants felt worse following low lottery outcomes for the partner if those outcomes were consequences of their own choice rather than the partner’s, which we interpret as interpersonal guilt.

BOLD signal

Next we sought to uncover the neural mechanisms associated with our guilt effect and those involved in tracking consequences of participants’ decisions on their partner. We analysed the BOLD responses of brain regions engaged during decision-making and at the time of receiving the outcomes of the choice using conventional as well as computational model-based analyses, using the fMRI data collected in Study 2.

Brain regions engaged during social decision-making

Given that our task involved several conditions and successive trial components, we aimed to first replicate previous results related to the neural correlates of decisions under economic risk. We searched for brain regions engaged more when participants chose the risky instead of the safe option, and found such responses in the bilateral ventral striatum (Cohen’s $d = 0.72$ and 0.85 in the left and right clusters, respectively; Figure 4A [↗](#) and Supplementary Table 3 [↗](#)), which replicates previous findings (Cui et al., 2022 [↗](#); Preuschoff et al., 2006 [↗](#)). Next, we aimed to identify the brain regions associated with social decision-making under risk, and searched for regions more engaged during decisions in the *Social* condition compared to the *Solo* condition. Three significant clusters of voxels were identified (Figure 4B [↗](#) and Supplementary Table 3 [↗](#)), in the precuneus ($d = 0.79$), the left temporo-parietal junction (TPJ; $d = 0.59$) and the medial prefrontal cortex (mPFC; $d = 0.54$). This also replicates previous findings associating this region with social decisions (e.g., Fareri et al., 2012 [↗](#); Jung et al., 2013 [↗](#); Nicolle et al., 2012 [↗](#); Ogawa et al., 2018 [↗](#); Piva et al., 2019 [↗](#)).

To assess whether activations in these five regions were sensitive to the interaction between choice and condition using a sensitive method, we analysed the responses in all the voxels in these regions using linear mixed models (LMMs). Our model included factors *Choice* (*Safe* or *Risky*), *Condition* (*Social*, *Partner* or *Solo*), *Run* (run one or two) and *ROI* (left and right ventral striatum, mPFC, precuneus and TPJ), with all interactions. We also tested the same model without the factor *Run*, but this fitted less well (higher BIC) and was thus discarded. As all factors and interactions were significant (data not shown for sake of brevity), we applied the same models to each region separately (Supplementary Table 4 [↗](#)). All regions showed a significant interaction between

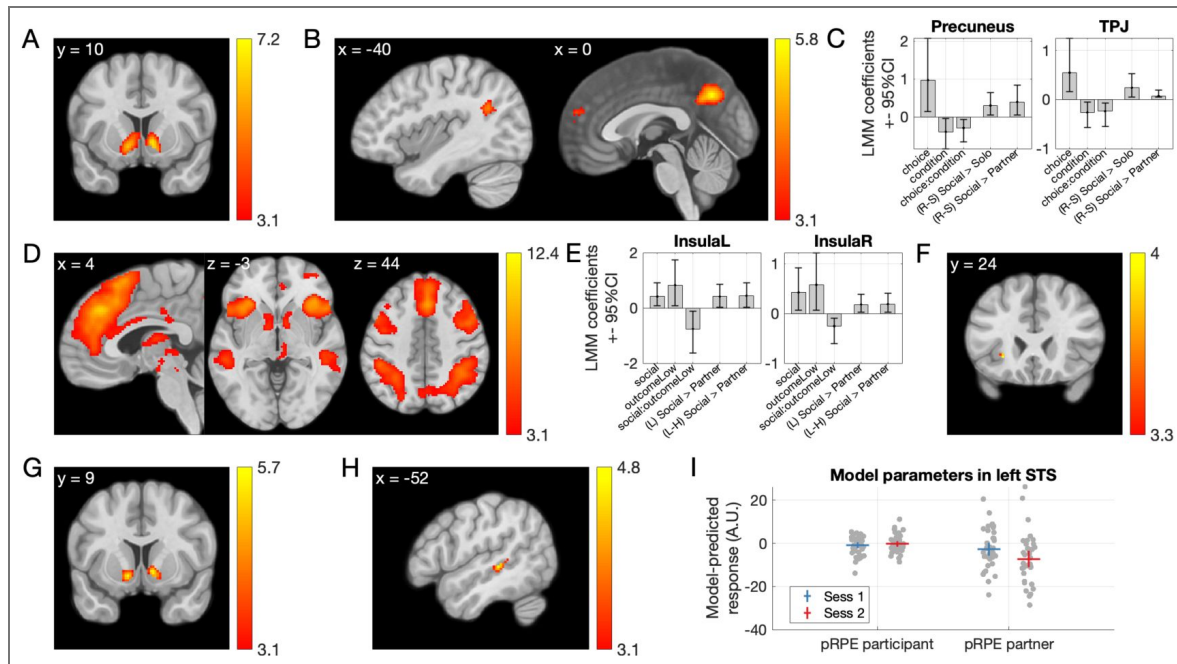


Figure 4. BOLD responses.

A. Regions showing a greater response when participants chose the risky (lottery) rather than the safe option, irrespective of Social or Solo condition. **B.** Regions showing a greater response when participants chose for both themselves and their partner rather than just for themselves (Social > Solo). **C.** Coefficients of linear mixed models indicate that two of these regions, precuneus and TPJ, were most active when participants chose the lottery in the Social condition. R-S indicates results of LMMs based on the Risky-Safe response difference. All coefficients and differences reported are significantly different from 0 (see Supplementary Table 4). **D.** Brain regions more active during receipt of the outcomes of lotteries than safe choices (all conditions). **E.** Coefficients of linear mixed models indicated that insula ROIs (see D) mirrored the guilt effect observed in our behavioural data: voxels here responded more to low lottery outcomes (L) for the partner when these resulted from participant's rather than the partner's choices, even when responses to high outcomes were subtracted (L-H). All coefficients and differences reported are significantly different from 0 (see Supplementary Table 6). **F.** A mass-univariate, voxel-wise analysis showed a compatible result: A cluster of voxels within the left insula ROI showed higher responses to low lottery outcomes for the partner if these resulted from participant rather than partner choices. **G.** Activation in bilateral ventral striatum explained by a computational model-based regressor coding participant rewards. **H.** Within brain regions sensitive to outcomes of risky choices, one cluster in the left superior temporal sulcus region showed a higher response to partner reward prediction errors resulting from participant rather than partner choices. **I.** Response in this cluster to the computational-model-based regressors coding participant reward prediction resulting from participant and partner choices, for both sessions of the experiment. All results shown survive thresholding at $p < 0.05$ corrected for multiple comparisons at the cluster level, based on a voxel-wise uncorrected threshold of $p < 0.001$. Colours in panels A-I indicate T values. In C and E, bars indicate the estimated coefficients. In C, E and I, error bars are 95% confidence intervals.

Condition and *Choice*, prompting us to examine these responses in detail. As we were particularly interested in the response to *Risky* decisions, we subtracted from it the responses to *Safe* decisions (labelled as “(R-S)” in Figure 4C) and compared this difference between (i) *Social* and *Solo* conditions, and (ii) between the *Social* and *Partner* conditions using LMMs. As above, we tested these models both with and without the factor *Run* and associated interaction, and we report the best-fitting model (see Supplementary Table 4). Only Precuneus and TPJ showed positive differences in both comparisons (Figure 4C), indicating that these regions were most active when participants chose the lottery in the *Social* condition, the critical situation in which participants assume responsibility over others.

Brain regions engaged during receipt of outcomes

Next, we focused on responses during choice outcomes. Cluster of voxels more active during receipt of lottery outcomes than outcomes of safe choices were identified in the bilateral anterior insula, dorsal mPFC (dmPFC), right superior temporal sulcus (STS), bilateral ventral striatum, right dorsolateral prefrontal cortex, and bilateral inferior parietal lobe (Figure 4D; for details including effect sizes, see Supplementary Table 5). Except for the STS, all these regions have been previously associated with processing of risk and/or ambiguity (Wu et al., 2021).

In our definition, guilt occurs due to responsibility for low lottery outcomes for the partner. To search for regions involved in this situation, we again analysed voxel responses using linear mixed models (LMMs). Our first model included factors *Lottery outcome* (*High* or *Low*), *Condition* (*Social* or *Partner*), *Run* (run one or two) and *ROI* (left and right ventral striatum, left and right insula, left and right parietal cortex, dmPFC, right prefrontal and right middle temporal regions), with all interactions. We also tested the same model without the factor *Run*, which fitted less well and was discarded. As in the analysis of responses to decisions above, all factors and interactions were significant (data not shown for sake of brevity), and we applied the same models to each region separately (Supplementary Table 6). To identify regions likely to be involved in the guilt effect, we selected those satisfying two conditions: higher activity in the *Social* compared to the *Partner* condition, and a significant *Social:LowOutcome* interaction. This procedure revealed the insulae (Figure 4E) and the right middle temporal cortex (trend significant *Social* vs *Partner* difference in the latter region). To test if these regions responded more to *Low* lottery outcomes in the *Social* condition, we ran additional models on the responses to *Low* lottery outcomes only (labelled as “(L) *Social* > *Partner*” in Figure 4E), and on this response minus the response to *High* lottery outcomes (labelled as “(L-H) *Social* > *Partner*” in Figure 4E); and indeed, the insulae showed the sought-after effect (see Supplementary Table 6 and Figure 4E). Thus, activation in our insula ROIs increased in situations during which participants experienced guilt for low outcomes impacting their partner, compared to similar outcomes resulting from the partner’s choices.

To attempt to confirm these results with a classic fine-grained mass-univariate voxel-wise analysis, we searched, within regions responding more to outcomes of risky compared to safe choices, for higher responses to low lottery outcomes for the partner following participant choices compared to the same outcomes resulting from partner choices. We found a weak response in a small cluster within the left anterior insula (peak $T = 3.95$, $d = 0.59$, 22 voxels, peak intensity at $[-28\ 24\ -4]$; Figure 4F). Given the documented association between anterior insula and guilt (see Introduction), we proceeded to test whether this result survived correction for family-wise errors due to multiple comparisons restricted to the left anterior insula gray matter [defined anatomically and thus independently from our findings, as the anterior short gyrus, middle short gyrus, and anterior inferior cortex in an anatomical maximum probability map (Faillenot et al., 2017)]. This correction resulted in a p value of 0.024. This result, although it is only a small effect in a small cluster, is consistent with the mixed model analysis reported earlier.

Neural correlates of responsibility for partner reward prediction errors revealed by computational model-based analysis

Next, we attempted to explain BOLD responses using predictions of the “*Responsibility*” computational model (see Behaviour / Computational modelling of happiness data, above). We used the model to create expected BOLD responses for each participant (see Methods) and as a

manipulation check searched for responses in ventral striatum evoked by participant rewards (O'Doherty et al., 2004, 2007). We found that activation in bilateral ventral striatum indeed increased with the amount of expected certain rewards and the expected values of chosen lotteries (left: $p_{FWE} = 0.002$, $T = 5.63$, $d = 0.75$, $Z = 5.41$, 110 voxels, peak at MNI [-14 8 -8], right: $p_{FWE} = 0.005$, $T = 5.46$, $d = 0.70$, $Z = 5.26$, 80 voxels, peak at MNI [10 10 -4], correction for multiple tests applied across the whole brain; Figure 4G). We thus used this model to search for voxels responding more to partner reward prediction errors resulting from participant rather than partner choices, within the regions sensitive to outcomes of risky choices. We found this effect in one cluster within the left STS ($p_{FWE} = 0.022$, $T = 4.70$, $d = 0.53$, $Z = 4.57$, 100 voxels, peak at MNI [-52 -32 0]; Figure 4H). Responses to the computational-model-based regressors coding for partner reward prediction errors resulting from participant and from partner choices in this cluster are shown in Figure 4I. This finding suggests that this region of the left STS tracks a partner's unexpected outcomes less when they do not follow from the participant's decisions.

Functional connectivity

Functional connectivity analyses have revealed differences in networks engaged by social and self-only choices (Jung et al., 2013; Ogawa et al., 2018), interactions between midbrain and anterior cingulate during compensation for guilt (Yu et al., 2014), and links between insula connectivity and responsibility aversion (Edelson et al., 2018). We hypothesized that connectivity with regions that showed guilt- and responsibility-related responses during the outcome phase (see previous paragraph) might change depending on whether participants made decisions for themselves only or for themselves and their partner, and depending on the type of choice (*Safe* or *Risky*). To this end, we used the left insula and left STS as seed regions for whole-brain seed-to-voxel psychophysiological interaction analyses (see Methods), to search for connectivity changes, during the choice phase of the trial, as a function of *Condition* and *Choice* (i.e., voxels with a significant interaction to *Condition* by *Choice*). The first analysis revealed a cluster in the right inferior frontal gyrus (IFG) whose connectivity to the insula (the seed region) was highest when participants made *Risky* choices for themselves and *Safe* choices for both players ($p_{FWE} = 0.020$, $T = 4.34$, $d = 0.80$, $Z = 4.21$, 115 voxels, peak at MNI [46 16 22]; Figure 5). A smaller cluster in the left IFG showed the same effect but did not survive correction for multiple tests ($p_{uncorrected} = 0.001$, $T = 4.08$, $Z = 3.97$, 38 voxels, peak at MNI [-38 -2 26]). The second analysis revealed a smaller cluster in the left inferior frontal gyrus that did not survive corrections for multiple tests, where connectivity with the left STS (the seed region) showed the opposite pattern: connectivity was highest when participants made *Safe* choices for themselves and *Risky* choices for both players ($p_{uncorrected} = 0.001$, $T = 4.44$, $Z = 4.30$, 35 voxels, peak at MNI [-48 14 6]; Supplementary Figure 2).

Comparison with multivariate neural guilt signature (Yu et al., 2020)

A recent study by Yu and colleagues re-analysed the results of two previous neuroimaging studies of guilt and obtained a neural multivariate guilt-related brain signature (guilt-related brain signature, GRBS) (Yu et al., 2020). As the GRBS and code to compare neural responses to it is freely available

(https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/Multivariate_signature_patterns/2019_Yu_Koban_Guilt), we compared this brain signature to the neural responses obtained in our task (response to low partner outcomes resulting from participant vs partner responses). The dot products between individual responses and the GRBS varied between -40.1 and 36.7, but overall these values were positive (mean = 5.22; median = 6.97; sign test: $p = 0.017$; Cliff's $\Delta = 0.4$ = medium effect size; data are not normally distributed). We assessed whether inter-individual differences in these dot product values correlated with the behavioural guilt responses, but did not find a significant association [Spearman's $Rho = -0.058$, $p = 0.725$].

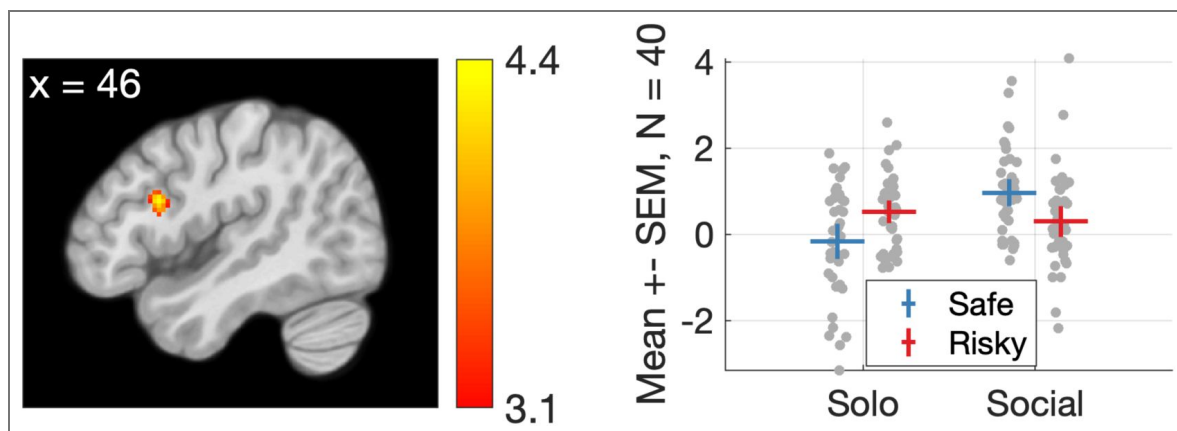


Figure 5. Changes in functional connectivity between the left anterior insula (seed) and a cluster in the right inferior frontal gyrus at the time of the choice as a function of condition (Social vs. Solo) and choice (Risky or Safe).

In the righthand panel, dots are data of individual participants, the markers represent means, and error bars indicate 95 confidence intervals about the mean.

Discussion

We report findings from an experiment on social responsibility and guilt in risky economic decisions, and their neural correlates. Being responsible for choosing a lottery that yielded a low outcome for a partner made our participants feel worse than witnessing the same outcome resulting from their partner's choice, which we interpret as interpersonal guilt; although we note that we have not asked participants specifically about which emotion they felt in these situations. Activation in the left anterior insula (aIns) reflected this effect, replicating previous associations between this region and feelings of guilt. Whole-brain activation patterns also resembled a neurometric marker of guilt (Yu et al., 2020 [↗](#)). Connectivity between aIns and the right inferior frontal gyrus varied depending on whether participants chose the risky or safe option and whether only the participant or both themselves and the partner were affected by the outcome of this choice, suggesting that this part of prefrontal cortex is sensitive to guilt-related information during social choices. Computational models explained trial-by-trial variations in momentary happiness during the task; the best-fitting model differentiated between partner reward prediction errors resulting from participant and partner choices, indicating that the impact of outcomes for the partner on participants' happiness varied depending on who chose. This confirms the importance of responsibility in determining the emotional consequences of risky social choices. fMRI analyses based on this computational model identified a left superior temporal sulcus (STS) region responding more to partner reward prediction errors resulting from participant rather than partner choices. This suggests a critical role of the STS in monitoring the consequences of one's risky decisions on others, an essential social cognitive function. A last analysis showed that connectivity between this STS region and the right Inferior Frontal Gyrus that varied depending on condition and choice, suggesting that the information processed in the STS is relayed to the IFG during social decisions. Our findings add to current understanding of the neural mechanisms underlying responsibility for, and guilt evoked by, the outcomes on others of social decisions under economic risk.

We used several approaches to compare choices made for self only or for both participant and partner. Participants made slightly more risk-seeking choices when deciding for themselves than for both themselves and the partner in Study 1, but this difference disappeared in Study 2. The ρ parameter on which this finding in Study 1 is based could only be estimated in a minority of participants due to a relatively low number of trials, which suggests that this finding may not be very reliable. The simpler and more robust method (evaluation of a risk premium) showed no difference in risk aversion across conditions in either study. Overall, we believe that we do not have strong evidence of differences in risk preferences across conditions.

Our findings are not unexpected given previous work. At least two studies reported that being responsible for somebody else's payoffs increases risk aversion (Fareri et al., 2022 [↗](#); Pahlke et al., 2015 [↗](#)). However, participants in a recent neuroimaging study were more loss averse when choosing for themselves rather than known other people (Arioli et al., 2023 [↗](#)). Indeed, recent large meta-analyses comparing risky choices for oneself vs. for others report either no difference in risk preferences (Batteux et al., 2019 [↗](#)) or a small shift towards more risky decisions for others (Polman & Wu, 2020 [↗](#)), with large variations across studies. Personal closeness to the others for which we decide seems to reduce differences in risk preferences (Fareri et al., 2022 [↗](#); Zhang et al., 2017 [↗](#)), as does making decisions for self before making decisions for others (Ifcher & Zarghamee, 2020 [↗](#)). In our study, decisions never only affected the partner, which most likely "watered down" any differences in risk attitude in decisions for oneself vs. for someone else. We also created a friendly prosocial environment in which people felt closer to each other than two strangers would (see Experiment Partner in Methods), and interleaved decisions for self only vs. for self and other. Similar risk preferences in decisions for self vs. self and other are thus unsurprising. The fact that participants made similarly risky decisions for themselves or for both during the fMRI study was to our advantage, because it made the neural signals evoked in these situations more comparable.

Anterior insula (aIns) response, particularly in the left hemisphere, was highest when partners received low lottery outcomes resulting from participants' risky choices. This replicates a large body of evidence associating aIns with feelings of guilt evoked during social decisions (see Introduction). Because we have neither asked our participants specifically what they felt in these situations, nor specifically whether they experienced guilt, we cannot exclude the possibility that they have instead or in addition felt empathy for their partner, a feeling of failure or bad luck, or some other emotion. The aIns region has also been associated with empathy for negative emotions such as disgust (Wicker et al., 2003 [↗](#)) or pain (Gu et al., 2012 [↗](#); Lamm et al., 2011 [↗](#)), with affective empathy during charitable giving (Tusche et al., 2016 [↗](#)), and more generally with emotion awareness (Bird et al., 2010 [↗](#); Gu et al., 2013 [↗](#)). These functions might be supported by interoception (Craig, 2002 [↗](#)): posterior insula is thought to receive interoceptive information from the body and to pass it on to the anterior insula for integration with sensory, emotional, cognitive, and motivational signals from other regions (Craig, 2009; Rogers-Carter & Christianson, 2019 [↗](#)). This integration would allow to represent one's own, as well as estimates of other people's, feelings and bodily states and allow error-based learning based on these (Lamm & Singer, 2010 [↗](#); Rogers-Carter & Christianson, 2019 [↗](#); Singer et al., 2009 [↗](#)). Our finding of changing functional connectivity between aIns and prefrontal cortex as a function of decision taken and experimental condition during choice may reflect changing emotional information transfer between these structures depending on choice and social context. How we feel when we witness our decisions' consequences on others is an important signal to consider when attempting to make good social decisions. Unsurprisingly, lesions of aIns are associated with reduced altruistic attitudes (Chau et al., 2018 [↗](#)), individuals with higher levels of psychopathic traits show reduced modulation of aIns response to anticipated guilt (Seara-Cardoso et al., 2016 [↗](#)), and show reduced guilt aversion (Gong et al., 2019 [↗](#)).

Deciding for both self and partner rather than just for oneself evoked increased activations in precuneus, left TPJ and dmPFC, areas classically associated with social cognition (Frith & Frith, 2006 [↗](#); Schurz et al., 2014 [↗](#); Van Overwalle, 2009 [↗](#)), and also with feelings of guilt (for meta-analyses, see Gifuni et al., 2017 [↗](#); Piretti et al., 2023 [↗](#)). This replicates previous findings of engagement of dmPFC, STS and/or TPJ when making decisions involving others (Jung et al., 2013 [↗](#); Nicolle et al., 2012 [↗](#); Ogawa et al., 2018 [↗](#); Piva et al., 2019 [↗](#)). Interestingly, our computational model-based analysis revealed a cluster of voxels in the left STS that responded positively to partner prediction errors, but only when these resulted from decisions made by the participant instead of the partner. This is congruent with associations between STS and cognitive perspective taking during charitable giving (Tusche et al., 2016 [↗](#)), representation of other people's interests during altruistic choice (Hutcherson et al., 2015 [↗](#)), or more generally with mentalizing computations during strategic social choice (Carter et al., 2012 [↗](#); Hampton et al., 2008 [↗](#); Hill et al., 2017 [↗](#)). The features of our experimental design (direct contrast of similar decisions made by the participant and partner; independent lotteries outcomes for self and other; quantification of the consequences of decisions through variations of momentary happiness) allowed to identify this very specific neural signal indicative of a neural sensitivity to the consequences of one's own actions on others, whether positive or negative. A neural signal coding partner reward prediction errors resulting from one's decisions seems essential to guide social decisions and as a basis for empathic concern for the people influenced by our actions.

Anecdotally, we found a weak functional connection between this left STS cluster and left aIns that varied as a function of choice and experimental condition; this is interesting because connections between a region in the left TPJ and aIns have been shown to vary depending on people's tendency to seek or avoid responsibility for others (Edelson et al., 2018 [↗](#)). Given the roles of the left STS/TPJ and aIns in guilt and social decisions, it is not surprising to find information exchanges between these structures. More experiments investigating specific computations in which these regions are involved (as in e.g., Charpentier & O'Doherty, 2018 [↗](#); Konovalov et al., 2021 [↗](#); Konovalov & Ruff, 2021 [↗](#)) are likely to help understand the neural mechanisms by which guilt and responsibility influence social decision-making.

There are several limitations to our study. The first limitation is that the partner's decisions were in fact taken by an algorithm. This was not communicated to the participants and was thus a case of deception, which is inadmissible in behavioural economics. To our defense, this study was planned as a social neuroscience study and was executed before we had taken into account the practices of behavioural economists. However, this approach did have the advantage to eliminate potentially complex iterative reciprocal influences of decisions and outcomes between the players, which would have led to much more complicated emotional states and decisions that would have been difficult to understand and model. Thus, from an analytical point of view, our approach might have actually allowed a cleaner comparison between decisions and outcomes than if the partner had really made the decisions. The fact that partner outcomes also influenced participants' momentary happiness demonstrates that participants were not emotionally detached from the consequences of their actions on their partner. This finding is of course essential for the validity of our results, and suggests that the effects could get stronger with more direct interactions. Therefore, although we will abstain from using deceptive practices when pursuing this research, we believe our findings to be valid.

Another limitation is that we have not asked participants to specifically name emotions as they proceeded through the experiment. As a result, we cannot ascertain whether people experienced guilt, shame, regret, disappointment or another specific emotion during the experiment. However, asking about specific emotions is itself associated with several drawbacks: emotion labels might have coloured participant's spontaneous emotional state; participant responses might be influenced by social desirability; thinking about which emotion best characterizes one's mental state takes time and distracts from the decision task; and the list of emotions to choose from is necessarily limited. In any case, the precise naming of emotions was not the aim of our study, instead we relied on a definition of guilt from psychological game theory (Battigalli & Dufwenberg, 2007 [\[4\]](#)). Indeed, previous work on social emotions recommends to “study the experiential phenomenology of emotions instead of mere emotion labels”, because “as a psychological explanation of human behaviour, the phenomenological experience of an emotion is much more important than the label attached to this experience” (Zeelenberg & Breugelmans, 2008 [\[4\]](#)). The “neurometrics” analysis of our guilt-related activation maps showed a significant similarity to a published guilt-related brain signature (Yu et al., 2020 [\[4\]](#)). We thus believe that our participants did experience guilt when their choices led to low outcomes for their partner.

Several open questions remain at the end of this study. As discussed above, asking participants directly about which emotions they have felt during the different stages of this task would allow us to link subjective experience with our analytical measures. Testing more participants would allow us to assess the impact of inter-individual variations in personality traits on the experience as well as the behavioural and neural correlates of guilt and responsibility.

Using more trials in the experiment would allow separate modelling of risk preferences in gain and loss trials in each experimental condition using expected utility models, and could allow testing whether changes in momentary happiness affect subsequent choices. Varying partner identities (friends, stranger, artificial agent) could reveal the impact of social discounting on guilt and responsibility. Functional connectivity analyses could also be performed for the data obtained in the feedback part of the paradigm. In sum, we believe that this experimental approach lends itself very well to the study of several aspects of social emotions.

Methods

Ethics

The studies fulfilled all relevant ethical regulations and were approved by the local ethics committee of the Medical Faculty of the University of Bonn, Germany (approval number 098/18). All participants gave written informed consent and the studies were conducted in accordance with the latest revision of the Declaration of Helsinki. Participants were remunerated for their time (10 Euros/hour) and received game earnings (0-10 Euros).

Participants

40 healthy participants (14 male, mean age 26.1, range 22 to 31) participated in Study 1 (behaviour only study), and 44 healthy participants (19 male, mean (SD) age = 30.6 (6.5), range 23 to 50) participated in Study 2 (fMRI study). All participants provided written informed consent. Participants were recruited from the local population through advertisements on online blackboards at the University of Bonn and on local community websites, and through flyers posted in libraries, university cafeterias and sports facilities. The number of participants recruited in Study 2 corresponds to the sample size estimated using G*Power 3.1 software (Faul et al., 2007) for a two-way t-test assessing the difference between two means (matched pairs) based on the results of Study 1 (Cohen's $d = 0.56$), with alpha error = 0.05 and power (1-beta) = 0.95, the required sample size was 44. The fMRI data from 4 participants in Study 2 were excluded from the fMRI data analysis because of excessive head motion ($>3\text{mm}$ or $>3^\circ$).

Experimental procedure

The design of the experiment was inspired by previous studies investigating how risky choices and their consequences influence momentary happiness (Rutledge et al., 2014, 2016). It was implemented in MATLAB (Version R2016b; *RRID:SCR_001622*; The MathWorks, Inc., Natick, MA) using the Psychtoolbox extensions (*RRID:SCR_002881*; <http://psychtoolbox.org>).

Experiment partner

Participants played with a friendly same-sex experiment partner (pairs of participants in Study 1, authors TW and MG in study 2), whom they met before the scan for an introduction session. In this session lasting about 15 minutes, participants played an ice breaker game with their experiment partner, in which both participant and partner took turns in drawing one half of a simple picture while being blindfolded and following verbal instructions given by their game partner. Encouragements and other positive feedback were given by the partner throughout the task. This icebreaker game led to an agreeable social atmosphere and positive, non-competitive, sympathetic attitude between the participants and the partner. Results of a brief questionnaire indicate that this approach was successful: participants' average ratings of their partners in terms of sympathy, cooperativity, honesty, openness and sociability were all above 8 on a scale of 1 to 10, in both studies (Supplementary Table 7).

Decision task

Following the icebreaker game, participants in Study 1 performed 3 sessions of a task in which they decided on each trial between a safe and a risky monetary option (Figure 1). The risky monetary option was a lottery with two equiprobable outcomes (lottery). There were 3 kinds of trials: decisions by the participant only for themselves (*Solo* condition), decisions by the participant for themselves and the partner (*Social* condition), and decisions by the partner for both themselves and the participant (*Partner* condition). Importantly, when the risky option was selected in the *Social* or *Partner* condition, the lottery was played out independently for the participant and the partner, such that both could receive the higher (*HighOutcome*) or lower outcome (*LowOutcome*), independently from each other. In order to ascertain constant decisions by the partner, the partner's decisions were simulated using a simple algorithm that always selected the option with the highest expected value; i.e., it selected the lottery if $EV_{diff} > 0$; $EV_{diff} = (high\ lottery\ outcome + low\ lottery\ outcome) / 2 - safe\ outcome$. The amounts earned per trial were determined as in a previous study (Rutledge et al., 2014), as follows. There were 20 mixed trials, 20 gain trials, and 20 loss trials per session. In the mixed trials, participants chose between a safe option of 0 € and a lottery consisting of a gain and a loss amount. Gain amounts were selected randomly out of the following: [15, 25, 40, 55, 75] cents. Loss options were determined by multiplying the gain amount with one randomly selected value out of the following: [-0.2, -0.34, -0.5, -0.64, -0.77, -0.89, -1, -1.1, -1.35, -2]. In gain trials, participants chose between a safe gain out of the following: [10, 15, 20, 25, 30] cents and a lottery with 0 € and a higher gain amount, calculated by multiplying the safe amount with one of the following values: [1.68, 1.82, 2, 2.22, 2.48, 2.8, 3.16, 3.6, 4.2, 5]. In loss trials, participants chose between a certain loss or a lottery with 0 € or a larger

loss. Certain loss amounts were one of [-10, -15, -20, -25, -30] cents, and the larger loss amount of the lottery was a multiple of the certain loss amount and one of the following values: [1.68, 1.82, 2, 2.22, 2.48, 2.8, 3.16, 3.6, 4.2, 5]. The position of the safe and risky option on the screen (left or right) was determined randomly on every trial, and the different trial types (*Solo*, *Social* or *Partner* condition; gain, loss and mixed trials) were presented in randomized order, with the constraint that there could not be more than 2 trials of the same condition in a row. Participants had unlimited time to choose between the safe or risky option. Decisions were displayed for 2s and lottery outcomes for 2.5s. Trials were separated in time by an inter-stimulus interval (ISI) of 1 to 2s drawn randomly from a gamma distribution.

Momentary happiness

Every two trials, one ISI after the outcome of the previous trial, participants were asked “How happy are you right now?”. They could respond by selecting a value on a scale from “very happy” (right) to “very unhappy” (left) by moving a cursor with a button press. The start position of the cursor was the midpoint of the scale, and the scale had 100 selectable options. For analysis, happiness ratings were Z-scored to cancel out effects of different rating variabilities across participants.

Difference between Studies 1 (behaviour only) and 2 (fMRI study)

In Study 2, participants performed 2 sessions of the experiment described above inside the fMRI scanner. All parameters were identical except that ISIs varied from 3 to 11s (drawn randomly from a gamma distribution). In Study 2, instead of playing against another same-sex participant, participants played either against experimenter MG or TW depending on sex (participant and partner were always of same sex) who were outside the scanner.

Statistical Analysis

1) General information

Statistical analysis was performed using Matlab R2024A (RRID:SCR_001622 [link](#)) and the lme4 package (Bates et al., 2022 [link](#)) (RRID:SCR_015654 [link](#), <https://cran.r-project.org/web/packages/lme4/index.html> [link](#)) in R (version 4.2.1, RRID:SCR_001905 [link](#)), and JASP (version 0.16.1, RRID:SCR_015823 [link](#), <https://jasp-stats.org/> [link](#), JASP Team 2021). fMRI data were analysed using SPM12 software (RRID:SCR_007037 [link](#); Wellcome Trust Centre for Neuroimaging, London, UK; <http://www.fil.ion.ucl.ac.uk/spm> [link](#)). All statistical tests were two-tailed. Bayes factors were calculated using default priors in JASP and express the probability of the data given H_1 relative to H_0 (BF_{10} , values larger than one are in favour of H_1). Effect sizes were calculated using standard approaches implemented in Matlab, including the Measures of Effect Size Toolbox (RRID:SCR_014703 [link](#), <https://www.mathworks.com/matlabcentral/fileexchange/32398-measures-of-effect-size-toolbox> [link](#)).

2) Decisions

a. Generalized linear mixed logistic regression

First, we assessed whether choices between the safe and risky options were reasonable and whether these choices varied between *Solo* and *Social* conditions. To do so, we modelled choices with a mixed-effects logistic regression with subject-specific random intercepts $u_{0j} \sim N(0, \delta_{u2})$ and slopes $[u_{1j}, u_{2j}] \sim N(0, \delta_{u2})$ and residual $\varepsilon_{ij} \sim L(0, 1)$, assumed to follow the standard logistic distribution. In this regression model, the choice of the risky option (Chose risky: 1 = Yes, 0 = No) was entered as the dependent variable, and explanatory variables were the difference between the expected value of the lottery minus the value of the safe option [$EV_{diff} = (high\ lottery\ outcome + low\ lottery\ outcome) / 2 - safe\ outcome$] and condition ($Cond$: 1 = *Solo*, 2 = *Social*; a categorical variable), as well as the interaction between them ($EV_{diff} \times Cond$). All explanatory variables were mean-centered. The random slopes are designed to account for between-subject heterogeneity:

$$Chose\ risky_{ij} = 1\ if\ \beta_0 + (\beta_1 + u_{1j}) * EV_{diff_{ij}} + (\beta_2 + u_{2j}) * Cond_{ij} + \beta_3 * EV_{diffXcond_{ij}} + u_{0j} + \varepsilon_{ij} > 0, \text{ and } Chose\ safe_{ij} = 0\ otherwise.$$

This gives rise to the regression equation:

$$\text{Chose risky}_{ij} = F(\beta_0 + (\beta_1 + u_{1j}) * \text{EVdiff}_{ij} + (\beta_2 + u_{2j}) * \text{Cond}_{ij} + \beta_3 * \text{EVdiff} * \text{Cond}_{ij} + u_{ij})$$

(Equation 1)

where $F(x) = 1 / [1 + \exp(-x)]$ is the cumulative distribution function of the standard logistic distribution. The subscript j indexes participants, while i indexes observations per participant. An observation corresponds to one choice (*Chose risky*: 1 = Yes, 0 = No).

This model was estimated for Study 1 and Study 2 separately, using the package *lme4* in R.

b. Estimating risk attitude 1: risk premium

Next, we estimated risk attitudes by calculating a *risk premium* for each participant and condition, defined as the difference between the expected value of a lottery and its certainty equivalent (positive values indicate risk aversion, negative values risk seeking). The certainty equivalent is the smallest amount an individual would be indifferent to spending on a lottery. We estimated the certainty equivalent of the lotteries by fitting to each participants' decisions in each condition the logistic regression model with residual $\varepsilon_{ij} \sim L(0, 1)$, assumed to follow the standard logistic distribution:

$$\text{Chose risky}_i = F(\beta_0 + \beta_1 * \text{EVdiff}_i)$$

(Equation 2)

The *risk premium* for each condition was defined as the EVdiff value corresponding to 50% *risky* choices indicated by the fitted logistic regression model. In both Studies, these *risk premium* values were compared between *Solo* and *Social* conditions using paired *t*-tests.

c. Estimating risk attitude 2: CARA model

Next, we estimated risk attitudes using expected utility theory. In this approach, an agent chooses between two options by comparing their expected utility values. Utility values reflect the agent's subjective values or preferences for a set of choice options, revealed by the agent's choices. Expected utility theory uses utility functions to describe the agents' preferences, including that agent's attitude towards risk. One simple model of risk aversion termed Constant Absolute Risk Aversion (CARA) assumes an exponential utility function:

$$U(x, \rho) = 1 - e(-\rho x)$$

Where x is a monetary amount and ρ is a constant that represents the degree of risk preference. U has an upward slope for perceived gains, indicating higher utility for a larger amount received. For a risk averse individual facing a choice in the domain of gains, ρ is positive and U is concave: a sure amount is preferred over a lottery with the same expected value. For a risk seeking individual facing the same choices, ρ is negative.

Under CARA, the expected utility of a lottery is:

$$EU_{\text{lottery}} = p * U(x_1, \rho) + (1-p) * U(x_2, \rho)$$

Where x_1 and x_2 are the two possible outcomes of the lottery and p is the probability of obtaining outcome x_1 .

The certainty equivalent of a lottery under CARA is:

$$CE_{\text{lottery}} = \log(1 - \rho * EU_{\text{lottery}}) / -\rho$$

For $\rho \neq 0$.

(Equation 3)

Following an econometric method (von Gaudecker et al., 2011), we estimated ρ by fitting a probit regression with a cumulative Gaussian link function Φ to choices obtained in the *Solo* and *Social* conditions of each participant:

$$\Phi = 0.5 * (1 + \text{erf}((CE - VS) / (\sqrt{2} * \sigma)));$$

where CE is the certainty equivalent of the lottery, VS the value of the safe alternative option, σ is Fechner noise and erf is the error function. The regression was fitted using nonlinear least-squares implemented in Matlab's `nlinfit.m`. In both Studies, these ρ values were compared between *Solo* and *Social* conditions using paired t -tests.

3) Momentary happiness

a. Computational models of happiness variations

Following previous work (Rutledge et al., 2014, 2016), we modelled the variations in momentary well-being (happiness ratings) using computational models that considered the influence of recent reward expectations, prediction errors resulting from those expectations, and differences in rewards obtained by the game partners. All models contained separate terms for certain rewards, expected value for lotteries and reward prediction errors, with influences that decayed exponentially over trials. We ran five models: i) a “Basic” model (Equation 4); the winning model in Rutledge et al. 2014 in which there was no *Social* condition) with certain rewards (CR), expected value of chosen lotteries (EV), participant's reward prediction errors ($sRPE$, where s stands for “self”) and a forgetting factor that makes events in more recent trials more influential than those in earlier trials (γ); ii) an “Inequality” model, in which one regressor modelled both disadvantageous and advantageous inequality (Equation 5); iii) a “Guilt-envy” model, the winning model in the social task in Rutledge et al., 2016, which contains separate additional regressors for disadvantageous and advantageous inequality (Equation 6); iv) a “Responsibility” model in which in addition to the participant's reward prediction errors ($sRPE$) we also modelled the partner's reward prediction errors ($pRPE$, where p stands for partner), with a distinction between $pRPE$ s resulting from participant choices ($social_pRPE$) and partner choices ($partner_pRPE$) (Equation 7); and v) a “Responsibility redux” model identical to the precedent except for the absence of $partner_pRPE$ s, as we realised that this regressor explained little variance (Equation 8). The Basic, Inequality and Guilt-envy models are identical to those used in Rutledge et al., 2016.

The equations of these models were as follows.

The equation for the Basic model was:

$$Happiness(t) = w_1 \sum_{j=1}^t \gamma^{t-j} CR_j + w_2 \sum_{j=1}^t \gamma^{t-j} EV_j + w_3 \sum_{j=1}^t \gamma^{t-j} sRPE_j$$

(Equation 4)

where t is trial number, γ is a forgetting factor ($0 \leq \gamma \leq 1$) that weighs events in more recent trials more heavily than events in earlier trials (exponential decay over time), and weights w_1 to w_3 capture the influence of the following different event types: CR_j is the certain reward received when the safe option was chosen instead of the lottery on trial j , EV_j is the average reward for the lottery if chosen on trial j and $sRPE_j$ is the participant's reward prediction error on trial j obtained as a result of choosing the lottery. Terms for unchosen options were set to zero. No constant term was used as ratings were z-scored, resulting in a mean of the data of 0.

As described above, we complemented the Basic model by adding terms representing additional influences on variations in participant happiness. Two models included regressors accounting for inequality, and two models included regressors accounting for partner reward prediction errors. The equation for the first of the inequality models, the “Inequality” model, was as follows:

$$\begin{aligned} \text{Happiness}(t) = & w_1 \sum_{j=1}^t \gamma^{t-j} CR_j + w_2 \sum_{j=1}^t \gamma^{t-j} EV_j + w_3 \sum_{j=1}^t \gamma^{t-j} sRPE_j \\ & + w_4 \sum_{j=1}^t \gamma^{t-j} \max(|S_j - P_j|, 0)_j \end{aligned}$$

(Equation 5)

Here, S_j and P_j are the rewards received by the participant and by their experiment partner, respectively, and thus w_4 relates to the influence of inequality of any kind on the variation in happiness. The equation of the “Guilt-envy” model was as follows:

$$\begin{aligned} \text{Happiness}(t) = & w_1 \sum_{j=1}^t \gamma^{t-j} CR_j + w_2 \sum_{j=1}^t \gamma^{t-j} EV_j + w_3 \sum_{j=1}^t \gamma^{t-j} sRPE_j \\ & + w_4 \sum_{j=1}^t \gamma^{t-j} \max(S_j - P_j, 0)_j + w_5 \sum_{j=1}^t \gamma^{t-j} \max(P_j - S_j, 0)_j \end{aligned}$$

(Equation 6)

Here, w_4 relates to advantageous inequality and w_5 relates to disadvantageous inequality, modelled separately. The next two models included terms relating to the partner’s RPE.

The equation for the “Responsibility” model was:

$$\begin{aligned} \text{Happiness}(t) = & w_1 \sum_{j=1}^t \gamma^{t-j} CR_j + w_2 \sum_{j=1}^t \gamma^{t-j} EV_j + w_3 \sum_{j=1}^t \gamma^{t-j} sRPE_j \\ & + w_4 \sum_{j=1}^t \gamma^{t-j} \text{social_pRPE}_j + w_5 \sum_{j=1}^t \gamma^{t-j} \text{partner_pRPE}_j \end{aligned}$$

(Equation 7)

In this model, the influence of the partner’s RPE resulting from participant choices in the *Social* condition (*social_pRPE*) is modelled separately from the influence of the partner’s RPE resulting from partner choices, in the *Partner* condition (*partner_pRPE*). The equation for the last model, “Responsibility redux”, was the same as for the Responsibility model (Equation 7), except that we omitted the regressor coding partner’s RPE resulting from partner choices, as they had a smaller impact on participant happiness. This led to the following equation:

$$\begin{aligned} \text{Happiness}(t) = & w_1 \sum_{j=1}^t \gamma^{t-j} CR_j + w_2 \sum_{j=1}^t \gamma^{t-j} EV_j + w_3 \sum_{j=1}^t \gamma^{t-j} sRPE_j \\ & + w_4 \sum_{j=1}^t \gamma^{t-j} \text{social_pRPE}_j \end{aligned}$$

(Equation 8)

Likelihood ratio test

To formally assess which of our models fitted the data best, we supplemented the AIC, BIC, R^2 and adjusted R^2 values reported in Table 1 with a series of *likelihood ratio tests*: we compared pairwise the likelihoods of the *Responsibility* model given the data to the likelihoods of all the other models. The equation for this test was the following:

$$LR = 2 * \log(L_o / L_i);$$

where L_0 and L_1 are respectively the likelihoods of the simpler and more complex model. The likelihood ratio follows a Chi-square distribution, with degrees of freedom equal to the number of extra parameters in the more complex model. The equation simplifies to:

$$LR = -2 * (\log(L_1) - \log(L_0))$$

(Equation 9)

Parameter recovery

Stability of the estimated parameters of these models was evaluated by attempting to recover parameters from synthetic data created using each participant's real estimated parameters. After fitting each participants' momentary happiness data (see above), we synthesized new momentary happiness data based on each participant's estimated parameters, added 1SD of noise to the happiness data, fitted the model to these synthetic data, and repeated this procedure 10 times, for both studies. We then compared these new estimated parameters to the actual parameters from which the synthetic data were generated, as follows: For each parameter, we calculated the mean of each participant's recovered parameters, and regressed these means on the participants' actual parameters (see Supplementary Figure 1 [↗](#)). The results show that the estimated parameters could be reliably recovered from noisy synthetic data.

b. Linear mixed model regressions of happiness variations

To better understand the impact of responsibility for outcomes of risky social choices on happiness, we focused on the happiness reported after outcomes of lottery choices in *Social* and *Partner* trials and evaluated the impact of who chose the lottery. We performed this analysis using a mixed-effects linear regression analysis with residual $\epsilon_{ij} \sim N(0, \sigma\epsilon^2)$ and subject-specific random intercepts $u_j \sim N(0, \sigma u^2)$. Specifically, we regressed Z-scored momentary happiness ratings reported after these outcomes on dummy variables representing occurrences (coded as 1 if event present, 0 otherwise) of the participant receiving the high lottery outcome, the partner receiving the high lottery outcome, the participant having chosen the lottery, and interactions between these terms.

$$\text{Happiness}(t) = 1 + \beta_1 sHigh_{ij} + \beta_2 pHigh_{ij} + \beta_3 sDec_{ij} + \beta_4 sHigh * pHigh_{ij} + \beta_5 sHigh * sDec_{ij} + \beta_6 pHigh * sDec_{ij} + u_j + \epsilon_{ij}.$$

(Equation 10)

sHigh and *pHigh* had a value of 1 if the participant or the partner (respectively) received the high lottery outcome and 0 otherwise, *sDec* had a value of 1 if the participant chose the lottery. All models were estimated for Study 1 and Study 2 separately, using the *lme4* package in R. In both studies, the model reported in Equation 1 [↗](#) fitted the data significantly better ($p < 2e-5$) than the simpler models without interactions (higher total and adjusted R^2 , see Supplementary Table 2 [↗](#)), but not significantly worse than the model with all interactions ($p > 0.5$; tested with the ANOVA function in R). The regressors of the different models tested and their relative effects are reported in Supplementary Table 2 [↗](#).

fMRI data acquisition and pre-processing

Imaging data were collected on a 3T Siemens TRIO MRI system (Siemens AG, Erlangen, Germany) with a Siemens 32-channel head coil. Functional data were acquired using a T2* echo-planar imaging (EPI) BOLD sequence, with a repetition time (TR) of 2500 ms, an echo time (TE) of 30 ms, 37 slices with voxel sizes of 2 x 2 x 3 mm, a flip angle of 90°, a field of view of 192 mm and PAT two acceleration. To exclude participants with apparent brain pathologies and facilitate normalisation of the functional data, a high-resolution T1-weighted structural image was acquired, with a TR of 1660 ms, a TE of 2540 ms, 208 slices with voxel sizes of 0.8 x 0.8 x 0.8 mm and a field of view of 256 mm. Data were then pre-processed and analysed as in previous studies (e.g., Schultz et al., 2019 [↗](#)).

using standard procedures in SPM12. The first five volumes of each functional time series were discarded to allow for T1 signal equilibration. The structural image of each participant was coregistered with the mean functional image of that participant.

Functional images were corrected for head movement between scans by a 6-parameter affine realignment to the first image of the time-series and then re-realigned to the mean of all images. The structural scan of each participant was spatially normalised to the current Montreal Neurological Institute template (MNI305) by segmentation and non-linear warping to reference tissue probability maps in MNI space, and the resulting normalisation parameters were applied to all functional images which were then resampled at 2 x 2 x 2 mm voxel size, then smoothed using an 8 mm full width at half maximum Gaussian kernel. Time series were de-trended by the application of a high-pass filter (cut-off period, 128 s).

fMRI data analysis

The pre-processed functional data were analysed using a standard two-stage approach based on the general linear model (GLM) implemented in SPM12 running in MATLAB, with corrections for multiple comparisons. Individual participants' data were modelled with fixed effects models, and their summary data were entered in random effects models for group statistics and inferences at the population level.

Fixed-effects models

The fixed effects models implemented a mass univariate analysis applied to the pre-processed data. The first GLM (GLM1) included the following event types for each session: presentation of choice options, safe choice, risky choice (=lottery), receipt of outcome of safe option, all modelled separately for the *Social*, *Partner* and *Solo* conditions; then receipt of high lottery outcome by the partner, separately coded for the *Social* and *Partner* conditions, and receipt of high lottery outcome by the participant; then receipt of low lottery outcome by the partner, separately coded for the *Social* and *Partner* conditions, and receipt of low lottery outcome by the participant; a final regressor modelled the button presses that occurred when reporting momentary happiness. All events were modelled as stick functions at the time of occurrence and convolved by SPM using the canonical hemodynamic function. In addition, we created another GLM (GLM2) with regressors designed to identify brain regions whose activation reflected the variables of the best-fitting computational model (see above). This model contained regressors coding the receipt of certain rewards (*CR*), the expected value of the chosen option (*EV*), the participant's reward prediction error (*sRPE*), the partner RPE resulting from decisions made by the participant (*social_pRPE*) and the partner RPE resulting from decisions made by the partner (*partner_pRPE*). Those regressors were applied as follows to each experimental session of each participant: 1) fitting the computational model to the momentary happiness data; 2) creation of a series of stick functions representing the time of occurrence of the modelled events during the scan, with stick magnitude determined by the fitted value calculated by the model; 3) convolution of the series of stick functions with the canonical BOLD function (as implemented in SPM). The 5 computational regressors were entered as covariates into each participant's model, which also contained one regressor modelling the button presses that occurred when reporting momentary happiness. All GLMs contained six motion-correction parameters included as regressors of no-interest to account for motion-related artifacts. Regression coefficients (parameter estimates) were estimated for each regressor in each voxel of each participant's brain.

Psychophysiological Interaction (PPI) analysis

We used the gPPI toolbox (McLaren et al., 2012 [DOI](#)) for SPM to run seed-to-voxel functional connectivity analyses. We ran two models for each participant, both based on GLM1, with the following seeds: the left insula cluster more sensitive to the outcomes of *Risky* vs. *Safe* choices, yielding GLM3 (see Figure 4D [DOI](#) and E [DOI](#)), and the left STS cluster responding more to *social_pRPE* than *partner_pRPE*, yielding GLM4 (Figure 4H [DOI](#)). Seeds were identical for all participants. PPI

models thus contained the same regressors as GLM1, plus the interaction between the first eigenvariate of the signal in the seed region and the regressors coding the experimental conditions of interest: *Safe* and *Risky* choice in *Social*, *Partner* and *Solo* conditions.

Random-effects models

Individual parameter estimate maps of the responses to the experimental conditions were assessed in random-effects tests to evaluate effects at the level of the group of participants. These tests were second-level models based on individual parameter estimate maps. We tested six models: a full factorial model to investigate the response during decisions [factors *Choice* (*Safe* or *Risky*) and *Condition* (*Social* or *Solo*), based on GLM1 estimate maps]; a full factorial model to investigate the response during the receipt of outcomes [factors *Outcome* (*Safe*, *HighOutcome* or *LowOutcome*) and *Condition* (*Social*, *Partner* or *Solo*), based on GLM1 estimate maps]; a paired t-test contrasting negative lottery outcomes for the partner resulting from participant or partner choices (based on GLM1 estimate maps); a model to investigate the response to the computational regressors (*CR*, *EV*, *sRPE*, *social_pRPE* and *social_prPE*, based on GLM2 estimate maps); and two full factorial models to investigate functional connectivity during decisions [factors *Choice* (*Safe* or *Risky*) and *Condition* (*Social* or *Solo*), based on estimate maps of GLM3 and GLM4].

Linear Mixed models

Similar to the behavioural data, we analysed GLM1 parameter estimates from voxels of specific regions of interest (see Results) using mixed-effects linear regressions with residual $\varepsilon_{ij} \sim N(0, \sigma\varepsilon^2)$ and subject-specific random effects $u_j \sim N(0, \sigma u^2)$. These models were implemented in Matlab and fitted with the `fitlme.m` function using restricted maximum likelihood estimation. We regressed parameter estimates on dummy variables representing variables of interest. For responses during the decision phase of the experiment, our models included factors *Choice* (*Safe* or *Risky*), *Condition* (*Social*, *Partner* or *Solo*), *Run* (run one or two) and *ROI*, with all interactions. For responses during the outcome phase, our models included the factors *Social* (value of 1 for the *Social* condition and 0 for the *Partner* condition), *LowOutcome* (value of 1 for *Low lottery outcome* and 0 for *High lottery outcome*), *Run*, with all interactions. Models without the factor *Run* were tested as well, and the best-fitting models were tested further. In all models, *Subject* was the only random factor (random intercept). To better understand significant interaction effects, we ran subsequent analyses on contrasts of parameter estimates (see Results and Supplementary materials).

Thresholding

All clusters reported survive a significance threshold of $p < 0.05$ with family-wise error correction for multiple comparisons (FWE) across the whole brain or a smaller volume when explicitly mentioned, based on an uncorrected voxel-wise (cluster-forming) threshold of $p < 0.001$.

Data availability

All behavioural data and group fMRI results maps needed to reproduce all the results and the figures of this study are freely available at <https://github.com/BonnSocialNeuroscienceUnit/ResponsibilityExperiment>. The entire raw dataset (anonymous data) is publicly available on OpenNeuro (<https://doi.org/10.18112/openneuro.ds005588.v1.0.0>). All the code used to replicate the experiment, analyse the data and produce the figures of this study are freely available at <https://github.com/BonnSocialNeuroscienceUnit/ResponsibilityExperiment>.

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Additional information

Author contributions

Maria Gädeke: Investigation, Data curation, Software, Methodology, Writing – original draft, Writing – review and editing; Tom Willems: Investigation, Data curation, Software, Methodology; Omar Salah Ahmed: Software, Methodology; Bernd Weber: Resources, Funding acquisition; René Hurlemann: Conceptualization, Resources, Funding acquisition; Johannes Schultz: Conceptualization, Methodology, Data curation, Formal analysis, Visualization, Software, Supervision, Project administration, Writing – original draft, Writing – review and editing.

Author ORCID iDs

Johannes Schultz: <https://orcid.org/0000-0003-4117-232X>

Additional files

[Supplementary Materials](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors aimed to characterize neurocomputational signals underlying interpersonal guilt and responsibility. Across two studies, one behavioral and one fMRI, participants made risky economic decisions for themselves or for themselves and a partner; they also experienced a condition in which the partners made decisions for themselves and the participant. The authors also assessed momentary happiness intermittently between choices in the task. Briefly, the results demonstrated that participants' self-reported happiness decreased after disadvantageous outcomes for themselves and when both they and their partner were affected; and this effect was exacerbated when participants were responsible for their partner's low outcome, rather than the opposite, reflecting experienced guilt. Consistent with previous work, BOLD signals in the insula correlated with experienced guilt and insula-right IFG connectivity was enhanced when participants made risky choices for themselves and safe choices for themselves and a partner.

Strengths:

This study implements an interesting approach to investigating guilt and responsibility; the paradigm in particular is well-suited to approach this question, offering participants the chance to make risky vs. safe choices that affect both themselves and others. I appreciate the assessment of happiness as a metric for assessing guilt across the different task/outcome conditions, as well as the implementation of both computational models and fMRI.

Weaknesses:

In spite of the overall strengths of the study, I think there are a few areas in which the paper fell a bit short and could be improved.

Comment on the revised submission:

I appreciate the authors' attention to all of my comments and questions regarding the initial version of the paper. However, I still do not believe that the point about the small volume correction in the insula has been adequately addressed. The authors claim that because the SVC was done using an anatomically defined ROI, that it is valid and not double dipping. I understand where the authors are coming from. However, there are a few issues here. First, any use of ROIs is best done via pre-registration (Gentili et al., 2021, *European Journal of Neuroscience*). Second, the whole set of analyses in this section leading up to the SVC seems somewhat circular. The first step was a whole brain contrast of lottery vs. safe outcomes, which revealed activation in many areas including the insula. Then, it appears that the parameter estimates from the insula were extracted and submitted offline to linear mixed models probing for effects of outcome magnitude, social condition and time, which revealed that the insula activation demonstrated the 'sought after' effect. Next, the manuscript states that the authors attempted to confirm these results with a univariate analysis for the so-called guilt effect within regions showing a stronger response to outcomes of risky relative to safe outcomes, which again showed activation in the insula (not surprisingly), and then a

small volume correction was applied to these insula voxels. While an anatomical ROI from a different study was used for the correction, the issue is that multiple analyses already revealed that the insula was involved in the effect of interest. It is unclear why this is even necessary given that the LMM analysis demonstrated the expected result.

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Reviewer #2 (Public review):

Summary

This manuscript focuses on the role of social responsibility and guilt in social decision making by integrating neuroimaging and computational modeling methods. Across two studies, participants completed a lottery task in which they made decisions for themselves or for a social partner. By measuring momentary happiness throughout the task, the authors show that being responsible for a partner's bad lottery outcome leads to decreased happiness compared to trials in which the participant was not responsible for their partner's bad outcome. At the neural level, this guilt effect was reflected in increased neural activity in the anterior insula, and altered functional connectivity between the insula and the inferior frontal gyrus. Using computational modeling, the authors show that trial by trial fluctuations in happiness were successfully captured by a model including participant and partner rewards and prediction errors (a 'responsibility' model), and model-based neuroimaging analyses suggested that prediction errors for the partner were tracked by the superior temporal sulcus. Taken together, these findings suggest that responsibility and interpersonal guilt influence social decision making.

Strengths

This manuscript investigates the concept of guilt in social decision making through both statistical and computational modeling. It integrates behavioral and neural data, providing a more comprehensive understanding of the psychological mechanisms. For the behavioral results, data from two different studies is included, and although minor differences are found between the two studies, the main findings remain consistent. The authors share all their code and materials, leading to transparency and reproducibility of their methods.

The manuscript is well-grounded in prior work. The task design is inspired by a large body of previous work on social decision making, and includes the necessary conditions to support their claims (i.e., Solo, Social, and Partner conditions). The computational models used in this study are inspired by previous work, and build on well-established economic theories of decision making. The research question and hypotheses clearly extend previous findings, and the more traditional univariate results align with prior work.

The authors conducted extensive analyses, as supported by the inclusion of different linear models and computational models described in the supplemental materials. Psychological concepts like risk preferences are defined and tested in different ways, and different types of analyses (e.g., univariate and multivariate neuroimaging analyses) are used to try to answer the research questions. The inclusion and comparison of different computational models provides compelling support for the claim that partner prediction errors indeed influence task behavior, as illustrated by the multiple model comparison metrics and the good model recovery.

The authors did a good job acknowledging other factors that could differ between the conditions, including the role of other emotions (like empathy) or agency in the decision making process. These additional analyses and nuances strengthen the manuscript and the interpretability of the findings.

Weaknesses

As the authors already note, they did not directly ask participants to report their feelings of guilt. The authors clearly describe this limitation, and also note that in addition to guilt, other emotions like empathy could also be at play in interpersonal decisions. Despite this limitation, this study provides insights into the neural and behavioral mechanisms of responsibility and guilt in social decision making, and how they influence behavior.

<https://doi.org/10.7554/eLife.105391.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary

The authors aimed to characterize neurocomputational signals underlying interpersonal guilt and responsibility. Across two studies, one behavioral and one fMRI, participants made risky economic decisions for themselves or for themselves and a partner; they also experienced a condition in which the partners made decisions for themselves and the participant. The authors also assessed momentary happiness intermittently between choices in the task. Briefly, results demonstrated that participants' self-reported happiness decreased after disadvantageous outcomes for themselves and when both they and their partner were affected; this effect was exacerbated when participants were responsible for their partner's low outcome, rather than the opposite, reflecting experienced guilt. Consistent with previous work, BOLD signals in the insula correlated with experienced guilt, and insula-right IFG connectivity was enhanced when participants made risky choices for themselves and safe choices for themselves and a partner.

Strengths:

This study implements an interesting approach to investigating guilt and responsibility; the paradigm in particular is well-suited to approach this question, offering participants the chance to make risky v. safe choices that affect both themselves and others. I appreciate the assessment of happiness as a metric for assessing guilt across the different task/outcome conditions, as well as the implementation of both computational models and fMRI.

We thank Reviewer 1 for their positive assessment of our manuscript.

Weaknesses:

In spite of the overall strengths of the study, I think there are a few areas in which the paper fell a bit short and could be improved.

We thank Reviewer 1 for their comments, which we have used to improve our manuscript. We hope that these changes address the issues raised by the Reviewer.

(1) While the framing and goal of this study was to investigate guilt and felt responsibility, the task implemented - a risky choice task with social conditions - has been conducted in similar ways in past research that were not addressed here. The novelty of this study would appear to be the additional happiness assessments, but it would be helpful to consider the changes noted in risk-taking behavior in the context of additional

studies that have investigated changes in risky economic choice in social contexts (e.g., Arioli et al., 2023 Cerebral Cortex; Fareri et al., 2022 Scientific Reports).

We certainly agree that several previously published studies have relied on risky choice tasks with social conditions. In this revised version, we now mention these two studies in the substantially revised Introduction.

(2) The authors note they assessed changes in risk preferences between social and solo conditions in two ways - by calculating a 'risk premium' and then by estimating rho from an expected utility model. I am curious why the authors took both approaches (this did not seem clearly justified, though I apologize if I missed it). Relatedly, in the expected utility approach, the authors report that since 'the number of these types of trials varied across participants', they 'only obtained reliable estimates for [gain and loss] trials in some participants' - in study 1, 22 participants had unreliable estimates and in study 2, 28 participants had unreliable estimates. Because of this, and because the task itself only had 20 gains, 20 losses, and 20 mixed gambles per condition, I wonder if the authors can comment on how interpretable these findings are in the Discussion. Other work investigating loss aversion has implemented larger numbers of trials to mitigate the potential for unreliable estimates (e.g., Sokol-Hessner et al., 2009).

We agree that we have not clearly justified why we have taken two approaches to assess risk preferences. In short, while the expected utility approach is a more comprehensive method to model a participant's choices, we had not sufficiently considered the need for the large number of trials required to fit such models when designing our experiment. Calculating the risk premium was the less comprehensive, simpler alternative that we could calculate for all participants. We have now mentioned this fact in the Results section. As the only difference in risk aversion across conditions was found in Study 1 using the expected utility method, which could only be successfully applied in a minority of participants, we believe that this difference should not be taken as a strong finding. We have now mentioned this fact in the revised Discussion.

(3) One thing seemingly not addressed in the Discussion is the fact that the behavioral effect did not replicate significantly in study 2.

We agree that we had not sufficiently discussed the fact that there were (slight but significant) differences in risk preferences between the Solo and Social conditions in Study 1 but not in Study 2. We now do so in the revised Discussion, and write the following:

"Participants made slightly more risk-seeking choices when deciding for themselves than for both themselves and the partner in Study 1, but this difference disappeared in Study 2. The ρ parameter on which this finding in Study 1 is based could only be estimated in a minority of participants due to a relatively low number of trials, which suggests that this finding may not be very reliable. The simpler and more robust method (evaluation of a risk premium) showed no difference in risk aversion across conditions in either study. Overall, we believe that we do not have strong evidence of differences in risk preferences across conditions."

(4) Regarding the computational models, the authors suggest that the Responsibility and Responsibility Redux models provided the best fit, but they are claiming this based on separate metrics (e.g., in study 1, the redux model had the lowest AIC, but the responsibility only model had the highest R^2 ; additionally, the basic model had the lowest BIC). I am wondering if the authors considered conducting a direct model comparison to statistically compare model fits.

We agree that we should run formal, direct model comparison tests. We now ran likelihood-ratio tests which showed that the Responsibility model was the best. We now report this in the Results section, just below Table 1:

“A likelihood ratio test (Equation 9) revealed that the Responsibility model fitted better than all the other models, including the *Responsibility Redux* model (Study 1: all $LR \geq 47.36$, $p < 0.0001$; Study 2: all $LR \geq 77.83$, $p < 0.0001$).”

(5) *In the reporting of imaging results, the authors report in a univariate analysis that a small cluster in the left anterior insula showed a stronger response to low outcomes for the partner as a result of participant choice rather than from partner choice. It then seems as though the authors performed small volume correction on this cluster to see whether it survived. If that is accurate, then I would suggest that this result be removed because it is not recommended to perform SVC where the volume is defined based on a result from the same whole-brain analysis (i.e., it should be done a priori).*

As indicated in the manuscript, the small insula cluster centered at $[-28\ 24\ -4]$ and shown in Figure 4F survived corrections for multiple tests within the anatomically-defined anterior insula (based on the anatomical maximum probability map described in Faillenot et al., 2017), which is independent of the result of our analysis. Functionally defining the small volume based on the same data would indeed be circular and misleading “double-dipping”. We have most certainly NOT done this. The reason why we selected the anterior insula is because it is one of the regions most frequently associated with guilt (see the explanations in our Introduction, which refers for example to Bastin et al., 2016; Lamm & Singer, 2010; Piretti et al., 2023). Thus we feel that performing small-volume correction within the anatomically-defined anterior insula is a valid analysis. We fully acknowledge that, independently of any correction, the effect and the cluster are small. We now write:

“We found a weak response in a small cluster within the left anterior insula (peak $T = 3.95$, $d = 0.59$, 22 voxels, peak intensity at $[-28\ 24\ -4]$; Figure 4F). Given the documented association between anterior insula and guilt (see Introduction), we proceeded to test whether this result survived correction for family-wise errors due to multiple comparisons restricted to the left anterior insula gray matter [defined anatomically and thus independently from our findings, as the anterior short gyrus, middle short gyrus, and anterior inferior cortex in an anatomical maximum probability map (Faillenot et al., 2017)]. This correction resulted in a p value of 0.024. This result, although it is only a small effect in a small cluster, is consistent with the mixed model analysis reported earlier.”

Reviewer #2 (Public review):

Summary

This manuscript focuses on the role of social responsibility and guilt in social decision-making by integrating neuroimaging and computational modeling methods. Across two studies, participants completed a lottery task in which they made decisions for themselves or for a social partner. By measuring momentary happiness throughout the task, the authors show that being responsible for a partner's bad lottery outcome leads to decreased happiness compared to trials in which the participant was not responsible for their partner's bad outcome. At the neural level, this guilt effect was reflected in increased neural activity in the anterior insula, and altered functional connectivity between the insula and the inferior frontal gyrus. Using computational modeling, the authors show that trial-by-trial fluctuations in happiness were successfully captured by a model including participant and partner rewards and prediction errors (a 'responsibility' model), and model-based neuroimaging analyses suggested that prediction errors for the partner were tracked by the superior temporal sulcus. Taken together, these findings suggest that responsibility and interpersonal guilt influence social decision-making.

Strengths

This manuscript investigates the concept of guilt in social decision-making through both statistical and computational modeling. It integrates behavioral and neural data, providing a more comprehensive understanding of the psychological mechanisms. For the behavioral results, data from two different studies is included, and although minor differences are found between the two studies, the main findings remain consistent. The authors share all their code and materials, leading to transparency and reproducibility of their methods.

The manuscript is well-grounded in prior work. The task design is inspired by a large body of previous work on social decision-making and includes the necessary conditions to support their claims (i.e., Solo, Social, and Partner conditions). The computational models used in this study are inspired by previous work and build on well-established economic theories of decision-making. The research question and hypotheses clearly extend previous findings, and the more traditional univariate results align with prior work.

The authors conducted extensive analyses, as supported by the inclusion of different linear models and computational models described in the supplemental materials. Psychological concepts like risk preferences are defined and tested in different ways, and different types of analyses (e.g., univariate and multivariate neuroimaging analyses) are used to try to answer the research questions. The inclusion and comparison of different computational models provide compelling support for the claim that partner prediction errors indeed influence task behavior, as illustrated by the multiple model comparison metrics and the good model recovery.

We thank the reviewer very much for their comprehensive description of our study and the positive assessment of our study and approach.

Weaknesses

As the authors already note, they did not directly ask participants to report their feelings of guilt. The decrease in happiness reported after a bad choice for a partner might thus be something else than guilt, for example, empathy or feelings of failure (not necessarily related to guilt towards the other person). Although the patterns of neural activity evoked during the task match with previously found patterns of guilt, there is no direct measure of guilt included in the task. This warrants caution in the interpretation of these findings as guilt per se.

We fully agree that not directly asking participants about feelings of guilt is a clear limitation of our study. While we already mention this in our Discussion, we have expanded our discussion of the consequences on the interpretation of our results along the lines described by the reviewer in the revised manuscript. We would like to thank the reviewer for proposing these lines of thought, and have now made the following changes to the text:

In the first paragraph of the discussion, we now write: “Being responsible for choosing a lottery that yielded a low outcome for a partner made our participants feel worse than witnessing the same outcome resulting from their partner’s choice, which we interpret as interpersonal guilt; although we note that we have not asked participants specifically about which emotion they felt in these situations.

Later on, in the third paragraph focusing on the anterior insula, we now write: “This replicates a large body of evidence associating aIns with feelings of guilt evoked during social decisions (see Introduction). Because we have neither asked our participants specifically what they felt in these situations, nor specifically whether they experienced guilt, we cannot exclude the possibility that they have instead or in addition felt empathy for their partner, a feeling of failure or bad luck, or some other emotion.”

As most comparisons contrast the social condition (making the decision for your partner) against either the partner condition (watching your partner make their decision) or the solo condition (making your own decision), an open question remains of how agency influences momentary happiness, independent of potential guilt. Other open questions relate to individual differences in interpersonal guilt, and how those might influence behavior.

How agency influences momentary happiness or variations thereof during the course of an experiment such as ours is an interesting question in itself. We now ran linear mixed models assessing agency (i.e. we compared happiness in conditions Solo & Social conditions vs. Partner condition), which revealed lower happiness in Solo and Social conditions (i.e. when it was the participant's turn to decide) in both studies. This is interesting in itself and may reflect the drive behind responsibility aversion reported by Edelson et al.'s 2018 study: being assigned the role of the decider in a social setting may make people slightly unhappy, perhaps due to "weight of the responsibility". We now report these findings in the Results section, including this proposed explanation; because we were not specifically interested in responsibility aversion, we do not discuss this further in the Discussion. The edited text is under the new subsection entitled 'Momentary happiness: effects of agency, responsibility and guilt', on page 12:

"Next, we assessed whether happiness varied depending on the participant's agency (Social + Solo vs. Partner), and found happiness to be lower when the participant chose, independent of the outcome (Study 1: $t(3600) = -3.92$, $p = 0.00009$, $\beta = -0.14$, 95% CI = [-0.20 -0.07]; Study 2: $t(2870) = -6.07$, $p = 0.000000001$, $\beta = -0.24$, 95% CI = [-0.31 -0.16]). . This is interesting in itself and may reflect the drive behind responsibility aversion reported by Edelson et al.'s 2018 study: being assigned the role of the decider in a social setting may make people slightly unhappy, perhaps due to "weight of the responsibility". To specifically search for a sign of interpersonal guilt, [...]"

Regarding individual differences: this is a very interesting topic that we have not addressed here due to the (relatively) small number of participants in our studies, but we might consider this for future follow-up studies, which we mention in the Discussion paragraph regarding open questions.

This manuscript is an impressive combination of multiple approaches, but how these different approaches relate to each other and how they can aid in answering slightly different questions is not very clearly described. The authors could improve this by more clearly describing the different methods and their added value in the introduction, and/or by including a paragraph on implications, open questions, and future work in the discussion.

We thank the reviewer for their appreciation of our complementary approach, and agree that we had not sufficiently explained the reasons why we used several methods. We have now added a paragraph explaining this at the end of the Introduction (page 5):

"We analysed our behavioural data using several complementary methods: choices were modelled with mixed-effects regressions serving as manipulation checks; risk preferences expressed in choices were assessed using a comprehensive expected utility model as well as with a simpler, more robust "risk premium" approach; and happiness data were fitted, in addition to the computational models, with several linear mixed models to assess the impact of both the participant's and their partner's rewards, the impact of agency and their interactions. Inspired by findings reported in previous neuroimaging of social emotions, we also used several methods to analyse our fMRI data, including conventional methods (both region-of-interest and mass univariate); mixed-effects regression models; computational model-based analyses (inspired by e.g. Konovalov et al., 2021; Rutledge et al., 2014); and

functional connectivity (e.g. Edelson et al., 2018; Kononov et al., 2021). The behavioural modelling is thus complemented by neuroimaging analyses that offer insight about both the activity in regions associated with guilt as well as their place in a wider network, providing an in-depth comprehensive analysis of the mechanisms behind guilt evoked by social responsibility.”

In addition, as suggested we added the following paragraph on open questions and future work in the Discussion:

“Several open questions remain at the end of this study. As discussed above, asking participants directly about which emotions they have felt during the different stages of this task would allow us to link subjective experience with our analytical measures. Testing more participants would allow us to assess the impact of inter-individual variations in personality traits on the experience as well as the behavioural and neural correlates of guilt and responsibility. Using more trials in the experiment would allow separate modelling of risk preferences in gain and loss trials in each experimental condition using expected utility models, and could allow testing whether changes in momentary happiness affect subsequent choices. Varying partner identities (friends, strangers, artificial agent) could reveal the impact of social discounting on guilt and responsibility. In sum, we believe that this experimental approach lends itself very well to the study of several aspects of social emotions.”

However, taken together, this study provides useful insights into the neural and behavioral mechanisms of responsibility and guilt in social decision-making and how they influence behavior.

We thank the reviewer again for their appreciation of our work and hope that our revisions improved the manuscript.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

The majority of my suggestions are in the public review, so I will not repeat them here. But in general, I like the paper, and in addition to my other comments, I think that there should be more discussion of the potential limitations of the study and conclusions that can be drawn. I also thought parts of the results were a little hard to follow, particularly in the 'momentary happiness' section. Perhaps an additional subsection here might help with flow.

We agree that we could have discussed further the limitations of our study and the conclusions that can be drawn from it, which we have now done in the last paragraphs of the Discussion in this revised version.

To improve the structure of the section on ‘momentary happiness’, we separated this section into two, entitled: ‘Momentary happiness: links to reward’ and ‘Momentary happiness: effects of agency, responsibility and guilt’, which should facilitate the reading of this long section. We proceeded in a similar manner for the Choices section, which is now subdivided into ‘Choices: manipulation check’ and ‘Choices: risk preferences’. We believe that these changes have indeed improved the readability of our manuscript.

Reviewer #2 (Recommendations for the authors):

Overall, I believe this manuscript was well-designed, consists of extensive analyses, and provides interesting new insights into the mechanisms underlying social decision-making. I mostly have some clarifying questions and minor comments, which are described below.

(1) Integration of prior findings in the first paragraphs of the Introduction. Although all the previous work described in the 2nd-5th paragraph introduction is interesting, it felt a bit like an enumeration of findings rather than an integrated introduction leading to the current research question. At the end of paragraph 5, it becomes clear how these findings relate to the current research question, but I believe it will improve the flow and readability of the introduction if this becomes clear earlier on.

We agree that we could have integrated the cited previous work into the Introduction so that the text builds up to the research question. We have now extensively reworked several paragraphs in the Introduction (pages 3-5) and hope that these changes have made it easier to follow.

(2) For the risk attitudes (Choices), you describe pooling the gains and losses and then comparing the social and solo conditions. I was wondering whether you also looked at potential differences between gains and losses (delta measure) for social versus the solo condition (so a comparison of the delta). Based on prior work, I can imagine that the difference in risk attitudes for gains and losses might differ when making decisions for yourself versus when you're doing it for a partner. In general, I was wondering how you explain these findings, as there is also a lot of work showing differences in risk-taking patterns for gains and losses.

We agree that we could have compared delta measures between solo and social conditions. However, as we describe in the Results section and comment on in the Discussion, the relatively low number of trials made separate fitting of gain and loss trials across conditions difficult. While this question could thus be addressed in subsequent versions of our experiment with more trials, such a fine-grained analysis of the decisions was not the focus of our current study.

(3) On page 11, you state: "in particular the partner's reward prediction errors resulting from the participants' decisions, i.e. those pRPE for which participants were responsible." From the results described in the paragraph above, this doesn't become clear (e.g., there's no distinction made between social_pRPE and partner_pRPE in the text), as it only discusses differences in weights between pRPE and sRPE. I would recommend including some more information in the main text on these main modeling findings, so one doesn't have to go to the Supplemental Materials to understand them.

We did indeed fail to report these findings in the text! We thank the reviewer for pointing this out. We have now edited this passage as follows:

“Crucially, we find here that the partner’s reward prediction errors (social_pRPE and partner_pRPE) contributed to explaining changes in participants’ momentary happiness: the Responsibility and ResponsibilityRedux models explained the data better than the models without these parameters (see Table 1). In particular, the partner’s reward prediction errors resulting from the participants’ decisions (social_pRPE), i.e. those pRPE for which participants were responsible, contributed to explaining our data (weights for social_pRPE were greater than 0: Responsibility model: Study 1: $Z = 2.85$, $p = 0.004$, Study 2: $Z = 3.26$, $p = 0.001$; Responsibility Redux model: Study 1: $Z = 2.93$, $p = 0.003$, Study 2: $Z = 3.30$, $p = 0.001$; weights for social_pRPE tended to be higher than weights for partner_pRPE: Responsibility model: Study 1: $Z = 2.14$, $p = 0.033$; Study 2: $Z = 1.41$, $p = 0.16$).”

(4) The functional connectivity findings seem to come out of nowhere and are not introduced or described anywhere prior in the manuscript. It is therefore not completely clear why you conducted these analyses, or what they add above and beyond previous analyses. Already introducing this method earlier on would fix that.

We agree that we could have introduced functional connectivity analyses earlier in the text, particularly given the many previous studies in our field using this technique. We have now done this at the end of a new last paragraph of the Introduction:

“Inspired by findings reported in previous neuroimaging of social emotions, we also used several methods to analyse our fMRI data, including conventional methods (both region-of-interest and mass univariate); mixed-effects regression models; computational model-based analyses (inspired by e.g. Konovalov et al., 2021; Rutledge et al., 2014); and functional connectivity (e.g. Edelson et al., 2018; Konovalov et al., 2021). The behavioural modelling is thus complemented by neuroimaging analyses that offer insight about both the activity in regions associated with guilt as well as their place in a wider network, providing an in-depth comprehensive analysis of the mechanisms behind guilt evoked by social responsibility.”

(5) For the functional connectivity findings: I was wondering why you only looked at the choice phase, and not at the feedback phase. I understand that previous work focused on the choice phase, but for the purpose of this study (focus on guilt), I can imagine it is also interesting to see what happens with feedback. In the discussion, you also state "How we feel when we witness our decisions' consequences on others is an important signal to consider when attempting to make good social decisions." (p. 19), which is more focused on the feedback rather than choice, and also supports the idea that looking at the feedback moment might be relevant.

We agree that we could also have looked at the functional connectivity during the feedback phase. The main reason why we had originally not done so was time constraints. At the current time we would in addition point out that the manuscript is already very long and contains many analyses of behavioural and fMRI data. Adding this analysis would cost additional time and would further delay the publication of our manuscript, which we would prefer to avoid. However, one could of course look at these effects in subsequent analyses of the same data or in subsequent versions of this experiment. We have now mentioned this in the Discussion, in the paragraphs on open questions.

Minor comments:

(1) For some of the Figures, it would be helpful if the subtitles were more informative. For Figure 2 and Figure 3 for example, it would be nice if Study 1 and Study 2 were not only mentioned in the figure description but also in the actual figure. For Figures 3 and 4, it would be helpful to have significance stars for the bar plots as well.

We agree that these changes make the figures more easily understandable and have implemented them all, except for adding stars on Figure 4, because all bar plots in panels C and E would have been labeled with two or more stars, which would have made the figure difficult to read. We have now mentioned the fact that all these coefficients were significant in the figure legend.

(2) For some of the Supplementary Results, it would be very helpful if there was a legend or description. This is already the case for most of the SR, but not for all.

We have now added a legend to all elements of the Supplementary Results.

Some questions that came to mind while going through them:

- Supplementary Table 1: which p-values correspond to the significance stars? This information is included for Supplementary Table 2, but not for ST1.

We have now added the missing information in ST1.

- Supplementary Figure 1: do the colors correspond to different participants?

We have now specified that the colors do indeed correspond to different participants.

- *Supplementary Table 5 (final table): what do the - represent? As in, why is there no value for "run" for the MPFC? At first, I thought you only included the significant values, but then I noticed a few non-significant values as well, so it wasn't completely clear to me why some of the values were missing. This also applies to Supplementary Table 6.*

We have indeed forgotten to explain this. The ‘-’ in Supplementary Tables 4 and 6 indicate that the linear mixed model without the factor ‘run’ was the better-fitting one. We have now added the following explanation in the text accompanying Supplementary Table 4:

“We tested these models both with and without the factor Run and associated interaction, and we report the best-fitting model in the table below: a dash (‘-’) in the row displaying parameters for the run and socialVsSolo:run regressors indicates that the model without factor run was better-fitting for this ROI.”

(3) *I came across a few minor typos or sentences that were not completely clear to me.*

- *On page 3: "Patients with damage to ventromedial prefrontal cortex (vmPFC) seem insensitive to guilt when playing social economic games (Krajchich et al., 2009)." This sentence felt a bit out of nowhere and doesn't logically follow from the previous sentences.*

We have now revised the descriptions of this previous study as well as several others and how they fit into the research question.

- *On page 3: "In another study, participant errors in a difficult perception task lead to a partner feeling pain and evoked activations in left aIns and dlPFC (Koban et al., 2013)." This sentence doesn't really flow, and from the wording, it is not completely clear whether it's the errors or the partner pain that led to the aIns and dlPFC activation.*

We have now revised the description of this study as well, as follows:

“In another study, partners received painful stimuli when participants made errors during a difficult perception task. These errors evoked activations in the left aIns and dlPFC in the participants (Koban et al., 2013).”

- *Supplementary Figure 1: there is a missing period after the sentence "We then compared these new estimated parameters to the actual parameters from which the synthetic data were generated"*

We have now added a missing comma after “generated”.

- *On page 5: "We ran two experiments, Study 1 outside fMRI and Study 2 during fMRI, with separate groups of participants." I would change "outside fMRI" to outside the MRI scanner or something like that, as it's not completely correct to say "outside fMRI".*

We have changed the sentence to “outside the MRI scanner”.

- *On page 6: for the first result, there are currently two p-values reported ($p < 2.5e-20$ and $p < 2e-16$). I believe this is an error?*

This was indeed an error! We have re-run this analysis, noticed that also the degrees of freedom were miscalculated, and have updated this result and the effect of condition (solo vs social). Results are almost identical as previously and all conclusions hold. We have also checked the other analyses reported in this paragraph – all results replicate exactly.

- *On page 6: "Supplemental Table 1" should be "Supplementary Table 1" (for consistency).*

Done.

| *On page 8: "participants in both conditions of both studies", I would change "of both studies" to "for both studies".*

Done.

| *On page 8: for the "Momentary Happiness" paragraph, it would be helpful if you could briefly describe the Rutledge method here, for people who are unfamiliar with the approach.*

We now write the following at the beginning of this paragraph:

“Following Rutledge and colleagues’ methodology, which considers that changes in momentary happiness in response to outcomes of a probabilistic reward task are explained by the combined influence of recent reward expectations and prediction errors arising from those expectations, we fitted computational models to each participant’s happiness data.”

| *On page 10: "Wilkoxon sign-rank tests", should be "Wilcoxon".*

Done.

We thank the reviewer for their careful reading of our manuscript. We believe that these changes have indeed improved our manuscript.

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